1. IIT-JEE syllabus

Law of mass action; equilibrium constant; exothermic and endothermic reactions; Le-Chatelier's Principle and its applications

2. Introduction

It is an experimental fact that most of the process including chemical reactions, when carried out in a closed vessel, do not go to completion. Under these conditions, a process starts by itself or by initiation, continues for some time at diminishing rate and ultimately appears to stop. The reactants may still be present but they do not appear to change into products any more.

A reaction is said to be reversible if the composition of reaction mixture on the approach of equilibrium at a given temperature is the same irrespective of the initial state of the system, i.e. irrespective of the fact whether we start with reactants or the products. Some examples are:

A	+	B	$\rightleftharpoons$	C	+	D
$H_2(g)$	+	$I_2(g)$	$\rightleftharpoons$	$2HI(g)$		
$Ag^+(aq)$	+	$Fe^{2+}(aq)$	$\rightleftharpoons$	$Fe^{3+}(aq)$	+	$Ag(s)$
$2NH_3(g)$			$\rightleftharpoons$	$N_2(g)$	+	$3H_2(g)$

2.1 Characteristics of Chemical Equilibrium

- The equilibrium is dynamic i.e. the reaction continues in both forward and reverse directions.
- The rate of forward reaction equals to the rate of reverse reaction.
- The observable properties of the system such as pressure, concentration, density remains invariant with time.
- The chemical equilibrium can be approached from either side.

A catalyst can hasten the approach of equilibrium but does not alter the state of equilibrium.

2.2 Types of Equilibria

There are mainly two types of equilibria

a) ***Homogeneous:*** Equilibrium is said to be homogeneous if reactants and products are in same phase.

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$
$CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$

b) ***Heterogeneous:*** Equilibrium is said to be heterogeneous if reactants and products are in different phases

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$
$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$
$NH_2CO_2NH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

2.3 Law of mass action

Guldberg and Waage proposed that "The rate at which a substance reacts is directly proportional to its active mass and rate of a chemical reaction is directly proportional to product of active masses of reactants each raised to a power equal to corresponding stoichiometric coefficient appearing in the balanced chemical equation".

For dilute solutions active mass is equal to concentration. Taking the example of the reaction.

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

We can write,

Rate of forward reaction $r_f \propto [N_2]\,[H_2]^3$ rate of reverse reaction $r_r \propto [NH_3]^2$

3. Equilibrium Constants, K_C & K_P

Let us consider a reaction of the type,

$$A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$$

The double headed arrow signifies that the reaction occurs in both the directions in measurable extent. What do you do at the start? You start with pure A & B which are the reactants, in a closed system. Initially A and B reacts very fast to give C and D. Now as soon as C and D are formed they also started giving back A and B. But A and B being present in greater quantity forces the reaction to occur in forward direction much faster than the backward one. So what is the net result? It's nothing but the net formation of C & D. In this way as the time passes, A & B decreases and hence their strength, on the other hand C & D increases and so is their strength. So ultimately a time comes when the forward reaction is balanced by backward reaction. This state is the equilibrium.

Now, before deriving the different related expression let us know what is Law of Mass Action. It states that,

"The rate of a reaction is directly proportional to the product of concentrations of reactants with the stoichiometric co-efficient being raised to the power"

e.g. for the reaction.

$$aA_{(g)} + bB_{(g)} \rightleftharpoons cC_{(g)} + dD_{(g)}$$

for the forward reaction,
$R_f \propto [A]^a [B]^b$
Where R_f denotes rate of forward reaction.
[A] denotes concentration of A
'a' , 'b' are the powers which are the coefficients of A & B respectively.
If k_f be the rate constant for forward reaction the expression can be rewritten as
$R_f = k_f[A]^a [B]^b$ …(1)
Similarly, for the backward reaction:
$R_b = k_b [C]^c [D]^d$ …(2)
At equilibrium:
Rate of forward reaction = Rate of backward reaction
From equation (1) and (2)
$R_f = R_b$
or $k_f [A]^a [B]^b = k_b [C]^c [D]^d$
Rearranging we get, $\frac{k_f}{k_b} = \frac{[C]^c[D]^d}{[A]^a[B]^b}$

Now, the expression on the L.H.S. is a constant. So it means the expression on the R.H.S. has to be a constant. In fact it is so for a particular reaction. But obviously at a given temperature. Effect of temperature we will study later. This constant is named as equilibrium constant and is denoted by K.

This implies that no matter what we start with (i.e., A & B or C&D or A+B+C or A+B+D or A+C+D or B+C+D or A+B+C+D) and how much of these we start with, the ratio of $\frac{[C]^c\,[D]^d}{[A]^a\,[B]^b}$ is a fixed quantity at a given temperature when the reaction reaches equilibrium. That is , if we assume that this reaction has K = 4 then no matter what we take initially and irrespective of how much we take, once equilibrium is reached the ratio of $\frac{[C]^c\,[D]^d}{[A]^a\,[B]^b}$ will always be equal to 4.

Now let us consider the following reaction

$A(s) + B(g) \rightleftharpoons C(s) + D(g)$

Its equilibrium constant , K would be ; K = $\frac{[C]\,[D]}{[A]\,[B]}$

Concentration of C is the number of moles of C per unit volume of solution. Concentration of D is the number of moles of D per unit volume of the container (we can assume that the gas occupies the entire container). The concentration of A is the number of moles of A per unit volume of A. The concentration of all solids and pure liquids is a constant. This is because if initially we take w gm of A, then the moles of A are w/M. The volume of A is w/d where d is the density of A. Therefore, the initial concentration of A is $\frac{w/M}{w/d} = \frac{d}{M}$. We can see that at equilibrium also the concentration of A remains as d/M (d and M are constants). In fact even if A were a pure liquid, its concentration would have remained constant.

Therefore, we bring all the constant terms on one side and we get

$$K[A] = \frac{[C][D]}{[B]}$$

This ratio which is a constant and which involves only those concentration terms which are variables is called K_c, the equilibrium constant in terms of concentration.

$$\therefore K_C = \frac{[C][D]}{[B]} = K\,[A]$$

Now, let us consider the reaction,

$$A(g) + 2B(g) \rightleftharpoons 3C\,(g) + 4D\,(g)$$

$$K_C = \frac{[C]^3[D]^4}{[A][B]^2}$$

We know that concentration of a gas can be expressed as P/RT as shown below:

$$PV = nRT$$

$\frac{P}{RT} = \frac{n}{V}$ = number of moles per litre = concentration

$$\therefore [C]=\frac{P_C}{RT}\text{ ; } [D] = \frac{P_D}{RT}\text{ , } [A]= \frac{P_A}{RT} \text{ and } [B] = \frac{P_B}{RT}$$

Where,

P_C = partial pressure of C
P_D = partial pressure of D
P_A = partial pressure of A
P_B = partial pressure of B

$$\therefore K_C = \frac{(P_C/RT)^3(P_D/RT)^4}{(P_A/RT)(P_B/RT)^2}$$

$$\Rightarrow \frac{P_C{}^3P_D{}^4}{P_AP_B{}^2} = K_C\,(RT)^{(3+4)-(1+2)}.$$

Since K_C is a constant and RT is also a constant, so the right hand side of the above expression is also a constant. This is called K_p, the equilibrium constant in terms of the partial pressure.

$$\therefore \frac{P_C{}^3P_D{}^4}{P_AP_B{}^2} = K_P$$

Therefore, for this particular equilibrium, the ratio of partial pressures is also a constant.

In general, the relation between K_P and K_C is $K_P = K_C\,(RT)^{\Delta n}$

Where Δn = number of moles of gaseous products – number of moles of gaseous reactants.

Now Δn can have three possibilities.

$$\Delta n < 0,\ \Delta n = 0,\ \Delta n > 0$$

Accordingly we can predict, out of Kp and Kc which one will be higher or lower. At $\Delta n = 0$ both Kp & Kc are the same.

Now let us assume that A is a solid or pure liquid. The changes now would be that K_C would look like this, $K_C = \frac{[C]^3[D]^4}{[B]^2}$ and following the above given sequence of derivation, K_p would like this

$$K_P = \frac{P_C^{\,3} P_D^{\,4}}{P_B^{\,2}}$$

Next we assume that A was a solute present in a solution then K_c would remain the same i.e, $K_C = \frac{[C]^3[D]^4}{[A][B]^2}$. Now if we try to express the concentrations in terms of partial pressures, we would fail to do that for A. It is not possible to express the concentration of a solution in terms of its pressure or vapour pressure and constants.

Therefore [A] remains as such

$$\therefore K_C = \frac{\left(\frac{P_C}{RT}\right)^3\left(\frac{P_D}{RT}\right)^4}{[A]\left(\frac{P_B}{RT}\right)^2}$$

$$\frac{P_C^{\,3} P_D^{\,4}}{[A]P_B^{\,2}} = K_C\,(RT)^{(3+4)-(2)}$$

The R.H.S. of the above expression is a constant which implies that the L.H.S is also a constant. This new expression cannot be called as either K_C or K_P since it contains both concentration terms and pressure terms. We call it K_{PC}. We can also see that if we take [A] to the R.H.S. the L.H.S. contains only pressure terms, but then it is not a constant since [A] is not a constant.

Therefore, we can conclude that for K_P to exist for a reaction it must fulfill two conditions:– (i) it must have at least one gas either in the reactants or in the products and (ii) it must not have any component in solution phase.

Illustration 1: ***Calculate K_c for the reaction $A_{(g)} + 2B_{(g)} \rightleftharpoons 3C_{(g)}$ when the reaction was started with 2 moles/litre of A and 3 moles/litre of B. The equilibrium concentration of C is 3 mole/litre.***

Solution:

	$A_{(g)}$	$+ 2B_{(g)} \rightleftharpoons$	$3C_{(g)}$
Initially molarity	2	3	0
Molarity at eqm.	2 – x	3–2x	3x

From question, at equilibrium, 3x = 3M

$\Rightarrow$ x = 1M

$\therefore$ [A] = 2 – x = 1

[B] = 1

$$\therefore K_c = \frac{[C]^3}{[A][B]^2} = \frac{(3)^3}{1\times(1)^2} = 27$$

Exercise 1: ***In a gaseous reaction of the type $A_{(g)} + 2B_{(g)} \rightleftharpoons 2C_{(g)} + D_{(g)}$, the initial concentration of B was 1.5 times that of A. At equilibrium the equilibrium concentration of A and D were equal. Calculate the equilibrium constant?***

3.1 Important relationships involving Equilibrium Constant

If we reverse an equation, K_c or K_p is inverted i.e.

If	$A_{(g)}$	+	$B_{(g)}$	$\rightleftharpoons$	$C_{(g)}$	+	$D_{(g)}$	$K_C = 10$
Then for	$C_{(g)}$	+	$D_{(g)}$	$\rightleftharpoons$	$A_{(g)}$	+	$B_{(g)}$	$K_C' = 10^{-1}$

If we multiply each of the coefficient in a balanced equation by a factor n, then equilibrium constant is raised to the same factor.

If $\frac{1}{2}N_2 + \frac{1}{2}O_2 \rightleftharpoons NO$ $\quad K_c = 5$

Then for $N_2 + O_2 \rightleftharpoons 2NO$ $\quad K_c' = 5^2 = 25$

If we divide each of the coefficients in a balanced equation by the factor n, then new equilibrium constant is nth root of the previous value.

If	$2SO_2$	+	O_2	$\rightleftharpoons$	$2SO_3$	$K_c = 25$
Then for	SO_2	+	$\frac{1}{2}O_2$	$\rightleftharpoons$	SO_3	$K_C' = \sqrt{25} = 5$

When we combine individual equation, we have to multiply their equilibrium constants for net reaction. If K_1, K_2 and K_3 are stepwise equilibrium constant for $A \rightleftharpoons B$, $B \rightleftharpoons C$, $C \rightleftharpoons D$. Then for $A \rightleftharpoons D$, equilibrium constant is $K = K_1 K_2 K_3$.

Significance of the Magnitude of on Equilibrium Constant:

- A very large value of K_C or K_P signifies that the forward reaction goes to completion or very nearly so.

- A very small value of K_C or K_P signifies that the forward reaction does not occur to any significant extent.

- A reaction is most likely to reach a state of equilibrium in which both reactants and products are present if the numerical value of K_c or K_P is neither very large nor very small.

Illustration 2: ***Calculate K_c for the reaction, $HI_{(g)} \rightleftharpoons \frac{1}{2} H_{2(g)} + I_{2(g)}$***

Given that the reaction $H_{2(g)} + I_2 \rightleftharpoons 2HI_{(g)}$, had been started with 1 mole of $H_{2(s)}$ and 3 mole of $I_{2(g)}$. At the equilibrium the moles of HI formed is 1.5 moles. The volume of the reaction vessel is 2 litre.

Solution:

	$H_{2(g)}$ +	$I_2 \rightleftharpoons$	$2HI_{(g)}$
Initially moles	2	3	0
Moles at eqm.	$2 - x$	$3 - x$	$2x$
Conc. of eqm.	$\frac{2-x}{2}$	$\frac{3-x}{2}$	$\frac{2x}{2}$

From question, $2x = 1.5$

$x = 0.75$

$\therefore [H_2] = \frac{2-x}{2} = \frac{1.25}{2}$

$[I_2] = \frac{3-x}{2} = \frac{2.25}{2}$

$[HI] = \frac{1.5}{2}$

$\therefore K_c$ for the reaction $= \frac{[HI]^2}{[H_2][I_2]}$

$$= \frac{\left(\frac{1.5}{2}\right)^2}{\frac{1.25}{1} \times \frac{2.25}{1}} = \frac{1.5 \times 1.5}{1.25 \times 2.25} = 0.8$$

Let the equilibrium constant K'_c for the reaction

$HI_{(g)} \rightleftharpoons \frac{1}{2} H_{2(g)} + \frac{1}{2} I_{2(g)}$

$$\therefore K'_c = \frac{1}{\sqrt{K_c}} = \frac{1}{\sqrt{\frac{1.5 \times 1.5}{1.25 \times 2.25}}} = \frac{1}{\sqrt{0.8}} = \frac{1}{0.89} = 1.124$$

Exercise 2: ***At a certain temperature, equilibrium constant (K_C) is 16 for the reaction:***

$SO_2(g) + NO_2(g) \rightleftharpoons SO_3(g) + NO(g)$

If we take one mole of each of the four gases in one litre container, what would be the equilibrium concentration of NO and NO_2?

3.2 The Reaction Quotient 'Q'

Consider the equilibrium

$PCl_5\ (g) \rightleftharpoons PCl_3(g) + Cl_2\ (g)$

At equilibrium $\frac{[Cl_2][PCl_3]}{[PCl_5]}$ = K_C. When the reaction is not at equilibrium this ratio is called 'Q_C' i.e., Q_C is the general term used for the above given ratio at any instant of time. And at equilibrium Q_C becomes K_C.

Similarly, $\frac{P_{Cl_2}P_{PCl_3}}{P_{PCl_5}}$ is called Q_P and at equilibrium it becomes K_P.

- If the reaction is at equilibrium, $Q = K_c$
- A net reaction proceeds from left to right (forward direction) if $Q < K_C$.
- A net reaction proceeds from right to left (the reverse direction) if $Q > K_c$

Illustration 3: ***For the reaction,***
$A_{(g)} + B_{(g)} \rightleftharpoons 2C_{(g)}$ at 25°C, at a 2 litre vessel contains 1, 2, 3 moles of respectively. Predict the direction of the reaction of
a) Kc for the reaction is 3
b) Kd for the reaction 6
c) Kc for the reaction

Solution: $A_{(g)} + B_{(g)} \rightleftharpoons 2C_{(g)}$

Reaction quotient Q = $\frac{[C]^2}{[A][B]} = \frac{\left(\frac{3}{2}\right)^2}{\frac{1}{2}\times\frac{2}{2}} = \frac{9}{2} = 4.5$

a) $\therefore Q < K_c$, therefore backward reaction will be followed

b) $\because Q < K_c$

$\because$ The forward reaction is followed

c) $Q = K_c$

$\therefore$ The reaction is at equilibrium

Exercise 3: ***The equilibrium constant is 0.403 at 1000K for the process***

$FeO_{(s)} + CO_{(g)} \rightleftharpoons Fe_{(s)} + CO_{2)g)}$

A steam of pure CO is passed over powdered FeO at 1000K until equilibrium is reached. What is the mole fraction of CO in the gas stream leaving the reaction zone?

4. The Le–Chatelier's Principle

This principle, which is based on the fundamentals of a stable equilibrium, states that "When a chemical reaction at equilibrium is subjected to any stress, then the equilibrium shifts in that direction in which the effect of the stress is reduced".

Confused with "stress". Well by stress here what I mean is any change of reaction conditions e.g. in temperature, pressure, concentration etc.

This statement will be explained by the following example.

Let us consider the reaction: $2NH_3\,(g) \underset{exo}{\overset{endo}{\rightleftharpoons}} N_2\,(g) + 3H_2\,(g)$

Let the moles of N_2, H_2 and NH_3 at equilibrium be a, b and c moles respectively. Since the reaction is at equilibrium,

$$\frac{P_{N_2} \times (P_{H_2})^3}{(P_{NH_3})^2} = K_P = \frac{(X_{N_2}.P_T)(X_{H_2}.P_T)^3}{(X_{NH_3}.P_T)^2}$$

Where,

X terms denote respective mole fractions and P_T is the total pressure of the system.

$$\Rightarrow \frac{\left(\frac{a}{a+b+c} \times P_T\right)\left(\frac{b}{a+b+c} \times P_T\right)^3}{\left(\frac{c}{a+b+c} \times P_T\right)^2} = K_P$$

Here, $\frac{a}{a+b+c}$ = mole fraction of N_2

$\frac{b}{a+b+c}$ = mole fraction of H_2

$\frac{c}{a+b+c}$ = mole fraction NH_3 $\Rightarrow \frac{ab^3}{c^2} \times \frac{(P_T)^2}{(a+b+c)^2}$ = K_P

Since $P_T = \frac{(a+b+c)RT}{V}$ (assuming all gases to be ideal)

$$\therefore \frac{ab^3}{c^2} \times \left(\frac{RT}{V}\right)^2 = K_P \qquad \ldots (1)$$

Now, let us examine the effect of change in certain parameters such as number of moles, pressure, temperature etc.

If we increase a or b, the left hand side expression becomes Q_P (as it is disturbed from equilibrium) and we can see that $Q_P > K_P$. The reaction therefore moves backward to make $Q_P = K_P$.

If we increase c, $Q_P < K_P$ and the reaction has to move forward to revert back to equilibrium.

If we increase the volume of the container (which amounts to decreasing the pressure), $Q_P < K_P$ and the reaction moves forward to attain equilibrium.

If we increase the pressure of the reaction then equilibrium shifts towards backward direction since in reactant side we have got 2 moles and on product side we have got 4 moles. So pressure is reduced in backward direction.

If temperature is increased the equilibrium will shift in forward direction since the forward reaction is endothermic and temperature is reduced in this direction.

However from the expression if we increase the temperature of the reaction, the left hand side increases (Q_P) and therefore does it mean that the reaction goes backward (since $Q_P > K_P$)?. Does this also mean that if the number of moles of reactant and product gases are equal, no change in the reaction is observed on the changing temperature (as T would not exist on the left hand side). The answer to these questions is No. This is because K_P also changes with temperature. Therefore, we need to know the effect of temperature on both Q_P and K_P to decide the course of the reaction.

Illustration 4: ***For the reaction***
$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$, $\Delta H = -$ xkcal
What will be the effect
a) When pressure is increased?
b) When temperature is increased?
c) What will be the effect on the equilibrium constant when temperature is increased?

Solution:
a) Forward reaction will be followed
b) Backward reaction will be followed
c) Since the reaction is exothermic, thereby on increasing temperature, backward reaction is followed. There K will decrease.

Exercise 4: ***A reaction vessel is charged with steam and ethene in a molar ratio of 10:1. Pressure is maintained at 1.0 MP_a and temperature at 400K. Assuming that the only process is $C_2H_{4(g)} + H_2O_{(g)} \rightleftharpoons C_2H_5OH_{(g)}$***
What fraction of ethene is converted to ethanol at equilibrium? K is 0.306.

4.1 Effect of Addition of Inert Gases to a Reaction at Equilibrium

1. ***Addition at constant pressure***

Let us take a general reaction
$aA_{(g)} + bB_{(g)} \rightleftharpoons cC_{(g)} + dD_{(g)}$

We know, $$K_p = \frac{\left(\frac{n_C}{\sum n} \times P_T\right)^c \left(\frac{n_D}{\sum n} \times P_T\right)^d}{\left(\frac{n_A}{\sum n} \times P_T\right)^a \left(\frac{n_B}{\sum n} \times P_T\right)^b}$$

Where,

n_C n_D, n_A, n_B denotes the no. of moles of respective components and PT is the total pressure and $\sum n$ = total no. of moles of reactants and products.

Now, rearranging,

$$K_P = \frac{n_c^c \times n_D^d}{n_A^a \times n_B^b} \times \left[\frac{P_T}{\sum n}\right]^{\Delta n}$$

Where $\Delta n = (c + d) - (a + b)$
Now, Δn can be = 0, < 0 or > 0
Lets take each case separately.

a) $\Delta n = 0$: No effect

b) Δn = '+ve' :

Addition of inert gas increases the $\sum n$ i.e. $\left(\frac{P_T}{\sum n}\right)$ is decreased and so is $\left(\frac{P_T}{\sum n}\right)^{\Delta n}$. So products have to increase and reactants have to decrease to maintain constancy of Kp. So the equilibrium moves forward.

c) Δn = '–ve' :

In this case $\left(\frac{P_T}{\sum n}\right)$ decreases but $\left(\frac{P_T}{\sum n}\right)^{\Delta n}$ increases. So products have to decrease and reactants have to increase to maintain constancy of Kp. So the equilibrium moves backward.

2. ***Addition at Constant Volume***

Since at constant volume, the pressure increases with addition of inert gas and at the same time $\sum n$ also increases, they almost counter balance each other. So $\left(\frac{P_T}{\sum n}\right)^{\Delta n}$ can be safely approximated as constant. Thus addition of inert gas has no effect at constant volume.

4.2 Dependence of K_P or K_c on Temperature

Now we will derive the dependence of K_P on temperature.

Starting with Arrhenius equation of rate constant

$$k_f = A_f e^{-E_{a_f}/RT} \qquad \ldots \text{(i)}$$

Where, k_f = rate constant for forward reaction, A_f = Arrhenius constant of forward reaction,

E_{a_f} = Energy of activation of forward reaction

$$k_r = A_r e^{-E_{a_r}/RT} \qquad \ldots \text{(ii)}$$

Dividing (2) by (3) we get,

$$\frac{k_f}{k_r} = \frac{A_f}{A_r} e^{\frac{\left(E_{a_r} - E_{a_f}\right)}{RT}}$$

We know that $\frac{k_f}{k_r} = K$ (equilibrium constant)

$$\therefore \; K = \frac{A_f}{A_r} e^{\frac{\left(Ea_r - Ea_f\right)}{RT}}$$

At temperature T_1

$$K_{T_1} = \frac{A_f}{A_r} e^{\frac{\left(E_{a_r} - E_{a_f}\right)}{RT_1}} \qquad \ldots \text{(iii)}$$

At temperature T_2

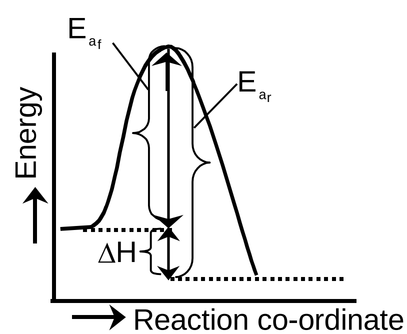
$$K_{T_2} = \frac{A_f}{A_r} e^{\frac{\left(E_{a_r} - E_{a_f}\right)}{RT_2}} \qquad \ldots \text{(iv)}$$

Dividing (iv) by (iii) we get

$$\frac{K_{T_2}}{K_{T_1}} = e^{\left(\frac{E_{ar} - E_{af}}{R}\right)\left(\frac{1}{T_2} - \frac{1}{T_1}\right)}$$

$$\Rightarrow \log \frac{K_{T_2}}{K_{T_1}} = \frac{E_{a_r} - E_{a_f}}{2.303\ R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

The enthalpy of a reaction is defined in terms of activation energies as $E_{a_f} - E_{a_r} = \Delta H$



$$\therefore \log \frac{K_{T_2}}{K_{T_1}} = \frac{-\Delta H}{2.303\ R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$\therefore \log \frac{\mathbf{K_{T_2}}}{\mathbf{K_{T_1}}} = \frac{\mathbf{\Delta H}}{\mathbf{2.303\ R}}\left(\frac{\mathbf{1}}{\mathbf{T_1}} - \frac{\mathbf{1}}{\mathbf{T_2}}\right) \qquad \ldots \text{(v)}$$

For an exothermic reaction , ΔH would be negative. If we increase the temperature of the system ($T_2 > T_1$), the right hand side of the equations (V) becomes negative.

$\therefore \; K_{T_2} < K_{T_1}$, that is, the equilibrium constant at the higher temperature would be less than that at the lower temperature. Now let us analyse our question. Will the reaction go forward or backward?

Before answering this, we must first encounter another problem. If temperature is increased, the new K_P would either increase or decrease or may remain same. Let us assume it increases.

Now, Q_P can also increase, decrease or remain unchanged. If K_P increases and Q_P decreases, than $Q_{P_{T_2}} < K_{P_{T_2}}$, therefore the reaction moves forward. If K_P increase and Q_P remains same, then $Q_{P_{T_2}}\left(Q_{P_{T_1}}\right) < K_{P_{T_2}}$. Again, the reaction moves forward. What, if K_P increase and Q_P also increases?

Will $Q_{P_{T_2}} = K_{P_{T_2}}$ or $Q_{P_{T_2}} < K_{P_{T_2}}$ or $Q_{P_{T_2}} > K_{P_{T_2}}$? This can be answered by simply looking at the dependence of Q_P and K_P on temperature. You can see from the equation (6) that K_P depends on temperature exponentially. While Q's dependence on T would be either to the power g,l,t…….. Therefore the variation in K_P due to T would be more than in Q_P due to T.

$\therefore$ K_P would still be greater than Q_P and the reaction moves forward again.

Therefore, to see the temperature effect, we need to look at K_P only. If it increases the reaction moves forward, if it decreases, reaction moves backward and if it remains fixed, then, no change at all.

Illustration 5: ***For the equilibrium***

$NH_4I_{(g)} \rightleftharpoons NH_{3(g)} + HI_{(g)}$

What will be the effect on the equilibrium constant on increasing the temperature.

Solution: Since the forward reaction is endothermic, so increasing the temperature, the forward reaction is favoured. Thereby the equilibrium constant will increase.

Illustration 6: ***K_p for the reaction $2BaO_{2(s)} \rightleftharpoons 2BaO_{(s)} + O_{2(g)}$ is 1.6×10^{-4} atm, at 400°C. Heat of reaction is – 25.14 kcal. What will be the no. of O_2 gas produced at 500°C temperature, if the is carried in 2 litre reaction vessel.***

Solution: We know that

$$\log\frac{K_{p_4}}{K_{p_1}} = \frac{\Delta H}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$\log\frac{K_{p_2}}{1.6\times 10^{-4}} = \frac{-25.14}{2.330\times 2\times 10^{-3}}\left[\frac{773-673}{773\times 673}\right]$$

$$\Rightarrow K_{p_2} = 1.46\times 10^{-5}\text{atm}$$

$$\therefore K_{p_2} = p_{o_2} = 1.46\times 10^{-5}\text{atm}$$

Since, PV = nRT

$$1.46\times 10^{-5}\times 2 = n\times 0.0821\times 773$$

$$\therefore n = \frac{1.46\times 10^{-5}\times 2}{0.0821\times 773}$$

Exercise 5: ***Equilibrium constant K_p for the reaction $\frac{3}{2}H_{2(g)} + \frac{1}{2}N_{2(g)} \rightleftharpoons NH_{3(g)}$ are 0.0266 and 0.0129 at 310°C and 400°C respectively. Calculate the heat of formation of gaseous ammonia.***

5. Degree of Dissociation (α)

Let us consider the reaction; $2NH_3\,(g) \rightleftharpoons N_2\,(g) + 3H_2\,(g)$

Let the initial moles of NH_3(g) be 'a'. Let x moles of NH_3 dissociate at equilibrium.

	$2NH_3$ (g)	$\rightleftharpoons$	N_2 (g)	+	$3H_2$ (g)
Initial moles	a		0		0
At equilibrium	a–x		$\frac{x}{2}$		$\frac{3x}{2}$

Degree of dissociation (α) of NH_3 is defined as the number of moles of NH_3 dissociated per mole of NH_3.

$\therefore$ if x moles dissociate from 'a' moles of NH_3, then, the degree of dissociation of NH_3 would be $\frac{x}{a}$.

We can also look at the reaction in the following manner.

	$2NH_3$ (g)	$\rightleftharpoons$	N_2 (g)	$+ 3H_2$ (g)
Initial moles	a		0	0
At equilibrium	a(1–α)		$\frac{a\alpha}{2}$	$\frac{3a\alpha}{2}$
or	a–2x′		x′	3 x′

where $\alpha = \frac{2x'}{a}$

Here total no. of moles at equilibrium is

$a - 2x' + x' + 3x' = a + 2x'$

Mole fraction of $NH_3 = \frac{a-2x'}{a+2x'}$

Mole fraction of $N_2 = \frac{x'}{a+2x'}$

Mole fraction of $H_2 = \frac{3x'}{a+2x'}$

The expression of K_p is

$$K_p = \frac{\left(\frac{x'}{a+2x'}\right)P_T \times \left(\frac{3x'}{a+2x'}\right)^3 \times P_T^3}{\left(\frac{a-2x'}{a+2x'}\right)^2 \times P_T^2} = \frac{27x'^4}{(a-2x')^2} \times \frac{P_T^2}{(a+2x')^2}$$

In this way you should calculate the basic equation. So my advice to you is that, while solving problem follow the method given below:

1. Write the balanced chemical reaction (mostly it will be given)
2. Under each component write the initial no. of moles.
3. Do the same for equilibrium condition.
4. Then derive the expression.

Do it and you are the winner.

5.1 Dependence of Degree of Dissociation on Density Measurements

The following is the method of calculating the degree of dissociation of a gas using vapour densities. This method is valid only for reactions whose K_P exist, i.e., reactions having at least one gas and having no solution.

Since PV = nRT

$$PV = \frac{w}{M}RT$$

$$M = \frac{wRT}{PV} = \frac{\rho RT}{P}$$

$$\therefore VD = \frac{\rho RT}{2P}$$

$$\text{Since } P = \frac{nRT}{V}$$

$$\therefore VD = \frac{\rho RT}{2nRT} \times V = \frac{\rho V}{2n}$$

For a reaction at eqb., V is a constant and ρ is a constant. $\therefore$ vapour Density $\alpha \frac{1}{n}$

$$\therefore \frac{\text{Total moles at equilibrium}}{\text{Initial total moles}} = \frac{\text{vapour density initial}}{\text{vapour density at equilibrium}} = \frac{D}{d} = \frac{M}{m}$$

($\because$ molecular weight = 2 $\times$ V.D)

Here M = molecular weight initial

m = molecular weight at equilibrium

Let us take a reaction

	PCl_5	$\rightleftharpoons$	PCl_3	+	Cl_2
Initial moles	C		0		0
At eqb.	C(1–α)		Cα		Cα

$$\therefore \frac{C(1+\alpha)}{C} = \frac{D}{d}; \quad 1+\alpha = \frac{D}{d} = \frac{M}{m}$$

Knowing D and d, α can be calculated and so for M and m.

Illustration 7: ***The degree of dissociation of PCl_5 is 66%, then find out the observed molar mass of the mixture.***

Solution:

	$PCl_{5(g)}$ $\rightleftharpoons$	$PCl_{3(g)}$ +	$Cl_{2(g)}$
Initially moles	1	0	0
At eqm.	$1-\alpha$	α	α

Where α = degree of dissociation = 0.6

Total moles at equilibrium = 1 + α = observed mole

$$\frac{\text{observed moles}}{\text{Theoretical moles}} = \frac{\text{molecular weight theoretical}}{\text{molecular weight observed}}$$

$$\Rightarrow \frac{1+\alpha}{1} = \frac{206.5}{\text{molecular weight observed}}$$

$$\Rightarrow \text{molecular weight observed} = \frac{206.5}{1+\alpha} = \frac{206.5}{1.6} = 129.06$$

Exercise 6: ***Vapour density of N_2O_4 which dissociates according to the equilibrium $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ is 25.67 at 100°C and a pressure of 1 atm. Calculate the degree of dissociation and K_p for the reaction.***

6. Relationship Between ΔG° and K

Free energy, G, denotes the self intrinsic electrostatic potential energy of a system. This means that in any molecule if we calculate the total electrostatic potential energy of all the charges due to all the other charges, we get what is called the free energy of the molecule. It tells about the stability of a molecule with respect to another molecule. Lesser the free energy of a molecule more stable it is.

Every reaction proceeds with a decrease in free energy. The free energy change in a process is expressed by ΔG. If it is negative, it means that products have lesser G than reactants, so the reaction goes forward. If it is positive the reaction goes reverse and if it is zero the reaction is at equilibrium.

ΔG is the free energy change at any given concentration of reactants and products. If all the reactants and products are taken at a concentration of 1 mole per liter, the free energy change of the reaction is called ΔG^o (standard free energy change).

One must understand that ΔG^o is not the free energy change at equilibrium. It is the free energy change when all the reactants and products are at a concentration of 1 mole/L. ΔG^o is related to K (equilibrium constant) by the relation, $\Delta G^o = -RT\ln K$. K may either be K_C or K_P. Accordingly we get ΔG_C^o or ΔG_P^o.

The units of ΔG^o depends only on RT. T is always in Kelvin, and if R is in Joules, ΔG^o will be in joules, and if R is calories then ΔG^o will be in calories.

Illustration 8: ***ΔG^0 for $\frac{1}{2}N_{2(g)} + \frac{3}{2}H_{2(g)} \rightleftharpoons NH_{3(g)}$ is – 16.5 kJ/mol. Find out K_p for the reaction. Also report K_p and ΔG^0 for $N_2 + 3H_2 \rightleftharpoons 2NH_3$ at 25°C.***

Solution:
$-\Delta G^0 = 2.303\ RT \log K_p$
$-(16.5 \times 10^3) = 2.303 \times 8.314 \times 298 \log K_p$
$\Rightarrow K_p = 779.4\ atm^{-1}$
Also k'_p for $N_2 + 3H_2 \rightleftharpoons 2NH_3$
$k_p^1 = (k_p)^2 = (779.4)^2$
$= 6.06 \times 10^5\ atm^{-2}$
Also $\Delta G^0 = 2.303\ RT \log k'_p$
$= 2.303 \times 8.314 \times 298 \log (6.07 \times 10^5) \Rightarrow -32.989$ kJ/mole

Exercise 7: ***The value of K_p at 298 K for the reaction $\frac{1}{2}N_2 + \frac{3}{2}H_2 \rightleftharpoons 2NH_3$ is found to be 826.0, partial pressure being measured in atm units calculate ΔG^0 at 298K.***

7. Solution to Exercises

Exercise 1: Given

$A_{(g)}$ +	$2B_{(g)}$	$\rightleftharpoons$	$2C_{(g)}$	+	$D_{(g)}$
a	$\frac{3}{2}a$		0		0
$(a - x)$	$\left(\frac{3}{2}a - 2x\right)$		$2x$		x

Given that $a - x = x$
or $(a = 2x)$
$K_C = \frac{(2x)^2 x}{x \times x} = 4$
$\therefore K_C = 4$

Exercise 2:

	$SO_2(g)$ +	$NO_2(g)$	$\rightleftharpoons$	$SO_3(g)$ +	$NO(g)$	$K_C = 16$
Initial (mole)	1	1		1	1	
At eq. (mole)	$1-\alpha$	$1-\alpha$		$1+\alpha$	$1+\alpha$	

$K_C = \left(\frac{1+\alpha}{1-\alpha}\right)^2 = 16$

$\frac{1+\alpha}{1-\alpha} = 4$

$\alpha = \frac{3}{5}$

$\therefore [NO] = 1 + \frac{3}{5} = \frac{8}{5} = 1.6$ M

$[NO_2] = 1 - \frac{3}{5} = \frac{2}{5} = 0.4$ M

Exercise 3: Let the initial moles of CO is 1 and x is the mole of CO reacted before equilibrium

$FeO_{(s)} + CO_{(g)} \rightleftharpoons Fe_{(s)} + CO_{2(g)}$

	1		0
	1 – x		x

Then given that $\frac{x}{1-x} = 0.403 \Rightarrow x = 0.287$

Mole fraction of remaining CO at equilibrium = $\frac{1-x}{1}$ = 1 – 0.287 = 0.713

Exercise 4:

C_2H_4	+	H_2O	$\rightleftharpoons$	C_2H_5OH	
1		10		0	at initial
1 – x		10 – x		x	at equilibrium

Total moles at equilibrium = 11 – x

$$\frac{\left(\frac{x}{11-x}\right)P}{\left(\frac{1-x}{11-x}\right)P\frac{(10-x)}{(11-x)}P} = K$$

$$\frac{x(11-x)}{(1-x)(10-x)P} = 0.306$$

$P =- \frac{10^6}{101.35} \Rightarrow x = 0.732 \therefore$ % dissociation = 73.2%

Exercise 5:

$$\log\frac{K_{P_1}}{K_{p_2}} = \frac{\Delta H}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$\log\frac{0.0129}{0.0266} = \frac{\Delta H}{2.303 \times 2}\left[\frac{1}{623} - \frac{1}{673}\right]$$

ΔH = – 12140 cals

Exercise 6:

$N_2O_4(g)$	$\rightleftharpoons$	$2NO_{2(g)}$	
1		0	at initial
1 – x		2x	at equilibrium

Total no. of moles at equilibrium = 1 + x

As we know that

$$\frac{V.D._{initial}}{V.D._{final}} = \frac{n_{final}}{n_{initial}}$$

$$\frac{46}{25.67} = \frac{1+x}{1} \Rightarrow x = 0.792$$

$$K_p = \frac{4x^2}{1-x^2} = \frac{4 \times (0.792)^2}{1-(0.792)^2} = 6.7 \text{ atm}$$

Exercise 7: $\Delta G^0 = 2.330 \text{ RT } \log K_p$

$= – 2.303 \times 1.98 \times 298 \times \log 826$

$\Rightarrow$ – 3980 calories

8. Solved Problems

8.1 Subjective

Problem 1: ***At 700K hydrogen and bromine react to form hydrogen bromide. The value of equilibrium constant for this reaction is 5 $\times$ 10^8^. Calculate the amount of H_2, Br_2 and HBr at the equilibrium if a mixture of 0.6 mole of H_2 and 0.2 mole of Br_2 is heated to 700K.***

Solution:

$$H_{2(g)} + Br_{2(g)} \rightleftharpoons 2HBr_{(g)}$$

Initially moles: 0.6, 0.2, 0

Conc. at eqbm.: $\frac{0.6-x}{V}$, $\frac{0.2-x}{V}$, $\frac{2x}{V}$

Now, where V = volume of container

$$\therefore K_{equilibrium} = \frac{[HBr]^2}{[H_2][Br_2]} = \frac{4x^2}{(0.6-x)(0.2-x)}$$

$$\text{or, } 5\times10^8 = \frac{4x^2}{(0.6-x)(0.2-x)}$$

x = 0.6 or, 0.2

x cannot be more than 0.2 when reaction is complete, some HBr will dissociation.

$$2HBr_{(g)} \rightleftharpoons H_{2(g)} + Br_{2(g)}$$

At equilibrium moles: 0.4 – 2x, 0.4 + x, x

Concn. at equilibrium: $\frac{0.4-2x}{V}$, $\frac{0.4+x}{V}$, $\frac{x}{V}$

$$\text{Here, } \frac{1}{K_{equilibrium}} = \frac{1}{5\times10^8} = \frac{x\times(0.4+x)}{(0.4-2x)^2}$$

$$\Rightarrow x = 2\times10^{-10}$$

$$\therefore \text{moles of } Br_2 = 2\times10^{-10}$$

moles of HBr = 0.4

moles of H_2 = 0.4

Problem 2: ***Anhydrous $CaCl_2$ is often used as dessicant. In presence of excess of $CaCl_2$, the amount of water taken up is governed by K_p = 1.28 $\times$ 10^{85} for the following reaction at room temperature,***

$CaCl_{2(s)} + 6H_2O_{(g)} \rightleftharpoons CaCl_2.6H_2O_{(s)}$

What is the equilibrium pressure of water in a closed vessel than contains $CaCl_{2(s)}$?

Solution:

$$CaCl_{2(s)} + 6H_2O_{(g)} \rightleftharpoons CaCl_2.6H_2O_{(s)}$$

$$\text{Here, } K_p = \frac{1}{\left(p_{H_2O}\right)} = 1.28\times10^{85}$$

$$\left(p_{H_2O}\right)^6 = \frac{1}{1.28}\times10^{-85}$$

$$\therefore p_{H_2O} = \left[\frac{1}{1.28}\times 10^{-85}\right]^{\frac{1}{6}}$$
$$p_{H_2O} = 6.54\times 10^{-15}\text{atm}$$

Problem 3: ***For the reaction***
$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$
K_p = 1.16 atm at 800°C. If 20g of $CaCO_3$ were kept in a 10 litre container and heated upt o800°C; what percentage by weight of $CaCO_3$ would remain unreacted at equilibrium?

Solution: For the reaction

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$
$$K_p = p_{CO_2} = 1.16\text{atm}$$
$$\therefore n_{CO_2} = \frac{p_{CO_2}\times V}{R\times T} = \frac{1.16\times 10}{0.0821\times 1073} = 0.132 \text{ mole}$$

Initial mole of $CaCO_3$ = $\frac{20}{100} = 0.2$

Mole of $CaCO_3$ reacted = 0.13

Moles of $CaCO_3$ unreacted = 0.2 – 0.132 = 0.168

$\therefore$ weight of $CaCO_3$ unreacted =0.068 $\times$ 100 = 6.8

% unreacted $CaCO_3$ by weight of $\frac{6.8}{20}\times 100$ = 3.4%

Problem 4: ***$NO_{2(g)} \rightleftharpoons NO_{(g)} + 1/2O_{2(g)}$***
Consider the above equilibrium where one mole of NO_2 is placed in a vessel and allowed to come to equilibrium at a total pressure of 1 atm. Experimental analysis shown that

Temperature	***700K***	***800K***
p_{NO}/p_{NO_2}	***0.872***	***2.50***

a) Calculate K_p at 700K and 800K
b) Calculate ΔG^0

Solution: a)

	$NO_{2(g)}$ $\rightleftharpoons$	$NO_{(g)}$ +	$\frac{1}{2}O_{2(g)}$
at t = 0	1 mole	0	0
t eq.	1 – 2x	2x	x
partial pressure at eqm.	$\left(\frac{1-2x}{1+2x}\right)P_T$	$\left(\frac{2x}{1+x}\right)P_T$	$\left(\frac{x}{1+x)}\right)P_T$

Given that

$$\frac{p_{NO}}{p_{NO_2}} = 0.872 \text{ at } 700K$$

$$\Rightarrow \frac{\left(\frac{2x}{1+x}\right)P_T}{\left(\frac{1-2x}{1+x}\right)P_T} = 0.872$$

$$\Rightarrow \frac{2x}{1-2x} = 0.872$$

$\Rightarrow$ x = 0.2329

$$\therefore\ p_{O_2} = \frac{x}{1+x}p_T = 0.1889$$

$$\therefore K_{p_1} = \frac{p_{NO}}{p_{NO_2}} \times \left(p_{O_2}\right)^{x/2}$$

$= 0.872 \times (0.1889)^{1/2}$

= 0.372 atm$^{1/2}$

Similarly K_{p_2} can be calculated

b) $\Delta G^0 = -RT\log K_p$

$= -2.303 \times 8.314 \times 700 \log 0.379$

= 5.65kJ/mol

Since,

$$\log\frac{K_{pr}}{K_{p_1}} = \frac{\Delta H}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$= \frac{\Delta H}{2.303 \times 8.314} \times \frac{100}{700 \times 800}$$

$$\log\frac{K_{pr}}{0.372} = \frac{\Delta H}{2.303 \times 8.314} \times \frac{100}{700 \times 800}$$

Problem 5: ***At 25° and 1 atm, N_2O_4 dissociates by the reaction***

$N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$

If it si 35% dissociated at given condition, find

i) The percent dissociation at same temperature if total pressure is 0.2 atm.

ii) The percent dissociation (at same temperature) in a 80g of N_2O_4 confined to a 7 litre vessel.

iii) What volume of above mixture will diffuse if 20 ml of pure O_2 diffuses in 10 minutes at same temperature and pressure?

Solution:

	$N_2O_{4(g)}$	$\rightleftharpoons$	$2NO_{2(g)}$
t equilibrium	$1 - \alpha$		2α
partial pressure at eqm	$\left[\frac{1-\alpha}{1+\alpha}\right]P_T$		$\left[\frac{2\alpha}{1+\alpha}\right]P_T$

$$K_p = \frac{p_{NO_2}^2}{p_{N_2O_4}} = \frac{4\alpha^2}{1-\alpha^2} \times p_T$$

$$= \frac{4 \times (0.35)^2}{1-(0.35)^2} \times 1 = 0.56 \text{ atm}$$

i) $K_p = \frac{4\alpha^2}{1-\alpha^2} \times P$

$0.56 = \frac{4\alpha^2}{1-\alpha^2} \times 0.2$

Solving, α = 09.64

% dissociation = 64%

ii) Since

$K_p = K_c(RT)^{\Delta n}$

Here Δn = 1

$\therefore K_c = \frac{K_p}{RT} = \frac{0.56}{0.0821 \times 298}$ = 0.0228 mole/litre

$\because K_c = \frac{4x\alpha^2}{(1-\alpha)V}$ where x = no. of moles of $N_2O_{4(g)}$

Neglecting α with respect to 1

i.e. $\alpha = \frac{80}{92}$

i.e. $1 - \alpha \approx \alpha$

$\therefore K_c = \frac{4x\alpha^2}{V}$

$0.0228 = \frac{4 \times \frac{80}{92}\alpha^2}{7} \Rightarrow \alpha = 0.2142$

$\therefore$ % dissociation $= 21.42\%$

iii) $\because 1 + \alpha = \frac{M_{th}}{M_{mix}}$

$1 + 0.2142 = \frac{92}{M_{mix}}$

$\therefore M_{mix} = \frac{92}{1.2142} = 75.76$

Let V(ml) volume of mixture diffused in Now, from Grahm's law of diffusion.

$\frac{r_{O_2}}{r_{mix}} = \sqrt{\frac{M_{mix}}{M_{O_2}}}$

$\frac{20/10}{V/10} = \sqrt{\frac{75.76}{32}}$

V = 12.998 ml

Problem 6: ***The value of K_C for the reaction: $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$ at 100°C is 50. If 1.0 L flask containing one mole of A_2 is connected with a 2.0 L flask containing two moles of B_2, how many moles of AB will be formed at 100°C?***

Solution: $A_{2(g)} + B_{2(g)} \rightleftharpoons 2AB_{(g)}$

As the two vessels are connected, the final volume for the contents is now 3.0 L. Let x mole each of A_2 and B_2 react to form 2x moles of AB_2 (from stoichiometry of reaction)

Moles	A_2	B_2	2AB
Initially	1	2	0
at equilibrium	1-x	2-x	2x

$K_C = \dfrac{[AB]^2}{[A_2][B_2]} = 50$; concentration of species at equilibrium are:

$[A_2] = (1 - x)/3$, $[B_2] = (2\text{-}x)/3$, $[AB] = 2x/3$

$$K_C = \frac{\left(\frac{2x}{3}\right)^2}{\left(\frac{1-x}{3}\right)\left(\frac{2-x}{3}\right)} = \frac{4x^2}{2-3x+x^2} = 50 \Rightarrow 46x^2 + 150x + 100 = 0$$

$\Rightarrow$ x = 0.93 or 2.33 (neglecting this value)

$\Rightarrow$ moles of AB(g) formed at equilibrium = 2x = 1.86

Problem 7: ***Solid NH_4I on rapid heating in a closed vessel at 357°C develops a constant pressure of 275 mm of Hg owing to the partial decomposition of NH_4I into NH_3 and HI but the pressure gradually increases further (when excess solid residually remains in the vessel) owing to the dissociation of HI. Calculate the final pressure developed under equilibrium.***

$NH_4I_{(s)} \rightleftharpoons NH_{3(g)} + HI_{(g)}$

$2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$ $\quad K_c = 0.015$ ***at 357°C***

Solution: $P = \dfrac{275}{760} = 0.3618$ atm

$p_{NH_3} = p_{HI} = \dfrac{0.3618}{2} = 0.181$ atm

$K_p = (0.181)^2 = 0.033\ atm^2$

$NH_4I_{(s)} \rightleftharpoons NH_{3(g)} + HI_{(g)}$

$(0.181 + p) \qquad (0.181 + p - p')$ $K_p = 3.3 \times 10^{-2}\ atm^2$

$2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$

$(0.181 + p - p') \qquad \dfrac{p'}{2} \qquad \dfrac{p'}{2} \qquad k_p = 0.015$

$0.033 = (0.018 - p)\ (0.181 + p - p')$

$0.015 = \dfrac{p'^2}{4(0.181 + p - p)^2}$

$\Rightarrow p' = 0.04$ atm

p = 0.026 atm

$\therefore$ total pressure = 2(0.181 + p) = 0.414 = 314.64 mm Hg

Problem 8: ***Two solids X and Y dissociates into gaseous products at a certain temperature as follows***

$X(s) \rightleftharpoons A(g) + C(g)$, and $Y(s) \rightleftharpoons B(g) + C(g)$. At a given temperature, pressure over excess solid X is 40 mm and total pressure over solid Y is 60 mm. Calculate.

i) the values of K_p for two reactions

ii) the ratio of mole sof A and B in the vapour state over a mixture of X and Y

iii) the total pressure of gases over a mixture of X and Y

Solution: $X(s) \rightleftharpoons A(g) + C(g)$ at equilibrium A & C are in equal proportions, so their pressures will be same.

Also $P_A + P_C = 40 \Rightarrow P_A = P_C = 20$ mm $\Rightarrow$ $K_p = P_A.P_C = 20^2 = 400$ mm^2

Similarly $Y(s) \rightleftharpoons B(g) + C(g)$

$P_B = P_C = 30$ mm $\Rightarrow K_p = P_B \,.\, P_C = 30^2 = 900$ mm^2

i) Now for a mixture of X and Y, we will have to consider both the equilibrium simultaneously.

$X(s) \rightleftharpoons A(g) + C(g)$ and

$Y(s) \rightleftharpoons B(g) + C(g)$

Let P_A = a mm, P_B = b mm

Note that the pressure of C due to dissociation of X will also be a mm and similarly the pressure of C due to dissociation of Y will also be b mm.

$\Rightarrow P_C = (a + b)$ mm

K_p (for X) $= P_A \,.\, P_C = a(a + b) = 400$ (I)

K_p (for Y) $= P_B.P_C = b(a + b) = 900$ (ii)

From (I) and (ii), we get

$$\frac{a}{b} = \frac{4}{9}$$

as volume and temperature are constant, the mole ratio will be same as the pressure ratio.

ii) The total pressure $= P_T = P_A + P_B + P_C$

= a + b + 9a + b) = 2(a + b)

to find a and b, solve the equations

$$\frac{a}{b} = \frac{4}{9} \text{ \& } a(a + b) = 400$$

$\Rightarrow$ a = 11.1 mm, b = 24.97 mm $\Rightarrow$ Total pressure = 2 (a + b) = 72.15

Problem 9: ***A saturated solution of I_2 in water contains 0.33 g of I_2 L^{-1}. More than this can be dissolved in a KI solution because of the following equilibrium :***

$I_2 + I^- \rightleftharpoons I_3^-$

A 0.1 M KI solution (0.1M I^-) actually dissolves 12.5 g I_2/ litre, most of which is converted to I_3^-, assuming that the concentration of I_2 in all saturated solution is the same, calculate the equilibrium constant (K_C) for the above reaction. What is the effect of adding water to a clear saturated solution of I_2 in the KI solution?

Solution:

	I_2 +	I^- $\rightleftharpoons$	I_3^-
Initial (conc.)	$\frac{12.5}{254}$	0.1	0
At eq. (conc.)	$\left(\frac{12.5}{254}-\alpha\right)$	$(0.1-\alpha)$	α

But $\frac{12.5}{254}-\alpha = \frac{0.33}{254}$

$0.0492 - \alpha = 0.0013$

$\alpha = 0.0479$

At eq. conc. (0.0013) 0.0521 0.0479

$\therefore K_C = \frac{(0.0479)}{(0.0013)(0.0521)}$

$K_C = 707.2$

Problem 10: ***0.15 mole of CO taken in a 2.5L flask is maintained at 750 K along with a catalyst so that the following reaction can take place.***

$CO(g)+2H_2(g) \rightleftharpoons CH_3OH(g)$

Hydrogen is introduced until the total pressure of the system is 8.5 atmosphere at equilibrium and 0.08 mole of methanol is formed. Calculate (a) K_P and K_C and (b) the final pressure if the same amount of CO and H_2 as before are used, but with no catalyst so that reaction does not take place.

Solution:

	$CO(g)$ +	$2H_2(g)$ $\rightleftharpoons$	$CH_3OH(g)$
Initial (mole)	0.15	a	0
At eq. (mole)	(0.15–x)	(a–2x)	x
	(0.15–0.08)	(a–0.16)	0.08

But, x = 0.08 mole = 0.07

Total number of moles at equilibrium = a – 0.01

number of moles at equilibrium = $\frac{PV}{RT} = \frac{8.5\times 2.5}{0.082\times 750} = 0.35$

$\therefore$ a– 0.01 = 0.35; a= 0.36

At equilibrium $[CO] = \frac{0.07}{2.5}$, $[H_2] = \frac{0.02}{2.5}$, $[CH_3OH] = \frac{0.08}{2.5}$

$$\therefore K_C = \frac{\left(\frac{0.08}{2.5}\right)}{\left(\frac{0.07}{2.5}\right)\left(\frac{0.02}{2.5}\right)^2}$$

$$K_C = \frac{0.08\times(2.5)^2}{0.07\times(0.02)^2} = 178.57\ M^{-2}$$

$\therefore K_P = 178.57\ (0.082\times 750)^{-2} = 0.047\ atm^{-2}$

(b) When no reaction takes place, so total moles = 0.51

$$\therefore P_{final} = \frac{nRT}{V}\ \frac{0.51\times 0.082\times 750}{2.5} = 12.546\ atm$$

Problem 11: ***For the equilibrium***

$LiCl.3NH_3$ (s) $\rightleftharpoons$ LiCl. NH_3 (s) + $2NH_3$,K_P = 9 atm^2 at 40°C. A five litre vessel contains 0.1 mole of $LiCl.NH_3$. How many minimum moles of NH_3 should be added to the flask at this temperature to derive the backward reaction for completion?

Solution:

$LiCl.3NH_3$ (s) $\rightleftharpoons$ $LiCl.NH_3$(s) + $2NH_3$(g)

K_P = 9 atm, Vol = 5l , Temperature = 40 °C

Number of moles of $LiCl.NH_3$ = 0.1

$2NH_3$(g)	+	$LiCl.NH_3$(s)	$\rightleftharpoons$	$LiCl.3NH_3$(s)
a–2x		0.1–x		x

0.1–x = 0 $\Rightarrow$ x = 0.1

a–2x = a–0.2

$K_P' = \frac{1}{K_P} = \frac{1}{9}$

$p_{NH_3} = \frac{(a-0.2)0.0821\times 313}{5}$ = 5.1395 (a–0.2)

$K_P' = \frac{1}{(5.1395)^2(a-0.2)^2}$

$(a-0.2)^2 = \frac{9}{(5.1395)^2}$ $\Rightarrow$ a = 0.7837

$\therefore$ Minimum number of moles of NH_3 required = 0.7837

Problem 12: ***For the gaseous reaction: $C_2H_2 + D_2O \rightleftharpoons C_2D_2 + H_2O$***

ΔH is 530 cal. At 25°C, K_P = 0.82. Calculate how much C_2D_2 will be formed if 1 mole of C_2H_2 and 2 moles of D_2O are put together at a total pressure of 1 atm at 100°C?

Solution:

Firstly calculate K_P at 100°C by using equation

$$\log \frac{(K_P)_{100^\circ C}}{(K_P)_{25^\circ C}} = \frac{\Delta H}{2.303R}\left(\frac{T_2 - T_1}{T_1T_2}\right)$$

On solving we get, $(K_P)_{100^\circ C}$ = 0.98

	C_2H_2	+	D_2O	$\rightleftharpoons$	C_2D_2	+	H_2O
Initial mole	1		2		0		0
At eq. mole	1–α		2–α		α		α

$\therefore 0.98 = \frac{\alpha^2}{(1-\alpha)(2-\alpha)}$

$0.02\alpha^2 + 2.94\alpha - 1.96 = 0$

On solving we get

α = 0.75

Problem 13: ***Solid Ammonium carbamate dissociates as:***

NH_2COONH_4(s) $\rightleftharpoons$ $2NH_3$(g) +CO_2(g). In a closed vessel solid ammonium carbamate is in equilibrium with its dissociation products. At equilibrium ammonia is added such that the partial pressure of NH_3 at new equilibrium now equals the original total pressure. Calculate the ratio of total pressure at new equilibrium to that of original total pressure.

Solution: $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

Let P = original equilibrium pressure, from the mole ratio of NH_3 and CO_2 at equilibrium, we have

$P_{NH_3} = \frac{2}{3}$ & $P_{CO_2} = \frac{P}{3}$

$\Rightarrow$ $K_p = (P_{NH_3})^2 \cdot P_{CO_2} = \left(\frac{2}{3}P\right)^2\left(\frac{P}{3}\right) = \frac{4}{27}P^3$

Now NH_3 is added such that, $P_{NH_3} = P$

Find the pressure of CO_2

$K_p = (P_{NH_3})^2 \cdot P_{CO_2} \Rightarrow \frac{4}{27}P^3 = P^2 \cdot P_{CO_2} \Rightarrow P_{CO_2} = \frac{4}{27}P$

Total new pressure = $P_{new} = P_{NH_3} + P_{CO_2}$

$\Rightarrow$ $P_{new} = P + \frac{4}{27}P = \frac{31}{27}P$

$\Rightarrow$ ratio = $\mathbf{\frac{P_{New}}{P_{original}} = \frac{27}{P} = \frac{31}{27}}$

Problem 14: ***For the reaction $NH_3\,(g) \rightleftharpoons \frac{1}{2}N_2\,(g) + \frac{3}{2}H_2\,(g)$***

Show that the degree of dissociation of NH_3 is given as

$\alpha = \left[1 + \frac{3\sqrt{3}}{4}\frac{P}{K_P}\right]^{-1/2}$

where P is the equilibrium pressure. If K_P of the above reaction is 78.1 atm at 400°C, determine the value of K_C.

Solution: Let n and α be the initial number of moles and degree of dissociation of $NH_{3(g)}$ respectively.

	$NH_3\,(g) \rightleftharpoons$	$\frac{1}{2}N_2\,(g)$ +	$\frac{3}{2}H_2\,(g)$
Initially	n	0	0
At equilibrium	$n(1-\alpha)$	$n\frac{\alpha}{2}$	$\frac{3n\alpha}{2}$

Total number of moles = $n(1+\alpha)$

Partial pressure of NH_3, $p_{NH_3} = \frac{1-\alpha}{1+\alpha} \times P$

Partial pressure of N_2, $p_{N_2} = \frac{\alpha}{2(1+\alpha)} \times P$

Partial pressure of H_2, $p_{H_2} = \frac{3\alpha}{2(1+\alpha)} \times P$

$\therefore K_P = \frac{(p_{N_2})^{1/2}(p_{H_2})^{3/2}}{p_{NH_3}}$

$$\frac{\left[\frac{\alpha P}{2(1+\alpha)}\right]^{1/.2}\left[\frac{3\alpha P}{2(1+\alpha)}\right]^{3/2}}{\frac{1-\alpha}{1+\alpha}\times P} = \left[\frac{3\sqrt{3}P}{4}\right]\left[\frac{\alpha^2}{1-\alpha^2}\right]$$

$$\frac{1-\alpha^2}{\alpha^2} = \frac{3\sqrt{3}P}{4\,K_P}$$

$$\frac{1}{\alpha^2} = \left[1+\frac{3\sqrt{3}}{4}\frac{P}{K_P}\right] \text{ or, } \alpha = \left[\mathbf{1}+\frac{\mathbf{3\sqrt{3}}}{\mathbf{4}}\frac{\mathbf{P}}{\mathbf{K_P}}\right]^{-1/2}$$

Δn, change in number of the moles of the given reaction = +1

$K_P = K_C(RT)^{\Delta n}$

or, $K_C = K_P\,(RT)^{-\Delta n}$

$K_C = 78.1\,[0.0821\times673]^{-1} = 1.413$ mol/l

Problem 15: ***In an evacuated vessel of capacity 110 litres, 4 moles of Argon and 5 moles of PCl_5 were introduced and equilibriated at a temperature at 250°C. At equilibrium, the total pressure of the mixture was found to be 4.678 atm. Calculate the degree of dissociation, α of PCl_5 and K_P for the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ at this temperature.***

Solution:

	$PCl_5(g)$ $\rightleftharpoons$	$PCl_3(g)$ +	$Cl_2(g)$
Initial moles	5	0	0
At equilibrium	5–x	x	x

Total moles = 5 + x + 4 (including moles of Argon)
= 9 + x

Since total moles $= \frac{PV}{RT} = \frac{4.678\times110}{0.082\times523}$ = 11.99 ≈ 12

$\therefore$ x = 3

$\therefore$ $\alpha = \frac{3}{5}$ = 0.6

$$K_P = \frac{\left(\frac{3}{12}\times4.678\right)^2}{\left(\frac{2}{12}\times4.678\right)} = 1.75 \text{ atm.}$$

The degree of dissociation of PCl_5 would have been 0.6 even in the absence of Argon. As one can see, the total pressure of gases constituting equilibrium is equal to 3.11 atm. The observed equilibrium pressure is 4.678 atm which means by the addition of 4 moles of Argon, the total pressure increases. This implies that addition of Argon has been done at constant volume which doesnot result in any change in degree of dissociation.

8.2 Objective

Problem 1: ***For which of the following K_p may be equal to 0.5 atm***

(A) $2HI \rightleftharpoons H_2 + I_2$ ***(B)*** $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

(C) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ ***(D)*** $2NO_2 \rightleftharpoons N_2O_4$

Solution: For K_p = 0.5 atm

$\Delta n = 1$ (since the unit is atm)

and $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

$\Delta n = 1$

$\therefore$ **(B)**

Problem 2: ***The vapour density of undecomposed N_2O_4 is 46. When heated, vapour density decreases to 24.5 due to its dissociation to NO_2. The % dissociation of N_2O_4 at the final temperature is***

(A) ***80*** ***(B)*** ***60***

(C) ***40*** ***(D)*** ***70***

Solution: $N_2O_4 \rightleftharpoons 2NO_2$

1	0	at initial
$1 - \alpha$	2α	at equilibrium

$$\therefore \frac{V.D._{innitial}}{V.D._{final}} = \frac{n_{final}}{n_{initial}}$$

$$\frac{46}{25.4} = \frac{25.4}{46} = \frac{1+\alpha}{1}$$

$1.8 = 1 + \alpha \Rightarrow \alpha = 0.8$

or 80%

$\therefore$ **(A)**

Problem 3: ***For the reaction $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$, the forward reaction at constant temperatured is favoured by***

(A) ***Introducing an inert gas at constant volume***

(B) ***Introducing Cl_2 gas at constant pressure***

(C) ***Introducing an inert gas at constant pressure***

(D) ***Increasing the volume of the container***

Solution: Introducing the inert gas at constant pressure will increase the volume of container and when the volume of container is increased equilibrium shift in the direction where no. of moles are increasing i.e. toward forward direction.

$\therefore$**(C) and (D)**

Problem 4: ***If pressure is applied to the following equilibrium, liquid $\rightleftharpoons$ vapours the boiling point of liquid***

(A) ***will increase*** ***(B)*** ***will decrease***

(C) ***may increase or decrease*** ***(D)*** ***will not change***

Solution: Boiling point of a liquid is the temperature at which vapour pressure became equal to atm pressure. If the pressure is applied to the above equilibrium the

reaction will go to the backward direction, i.e. vapour pressure decrease hence the boiling point increase.
$\therefore$ **(A)**

Problem 5: ***For the reaction***
$A_{(g)} + B_{(g)} \rightleftharpoons 3C_{(g)}$ at 250°C, a 3 litre vessel contains 1, 2, 4 mole of A, B and C respectively. If K_C for the reaction is 10, the reaction will proceed in
(A) Forward direction ***(B) Backward direction***
(C) In either direction ***(D) In equilibrium***

Solution:
$$Q = \frac{[C]^3}{[A][B]} = \frac{4^3 \times 3 \times 3}{3^3 \times 1 \times 2} = 10.66$$
$\because$ [C] = $\frac{4}{3}$

[A] = $\frac{1}{3}$

[B] = $\left(\frac{2}{3}\right)$ $\because$ K_C = 10, and Q $>$ K_C

$\therefore$ reaction will proceed in backward direction
$\therefore$ **(B)**

Problem 6: ***The equilibrium constant for the reaction $N_{2(s)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ is 4×10^{-4} at 200 K. In presence of a catalyst the equilibrium is attained 10 times faster. Therefore, the equilibrium constant in presence of catalyst at 200K is***
(A) 40×10^{-4} ***(B) 4×10^{-4}***
(C) 4×10^{-3} ***(D) Can't be calculated***

Solution: K_p and K_C value doesn't depend on catalyst
$\therefore$**(B)**

Problem 7: ***In a system $A_{(s)} \rightleftharpoons 2B_{(g)} + 3C_{(g)}$. If the concentration of C at equilibrium is increased by a factor 2, it will cause the equilibrium concentrations of B to change to***
(A) Two times of its original value
(B) One half of its original value
(C) $2\sqrt{2}$ time of its original value
(D) $\frac{1}{2\sqrt{2}}$ times of its original value

Solution: $A_{(s)} \rightleftharpoons 2B_{(g)} + 3C_{(g)}$
$K_C = [C]^3 [B]^2$
If (C) becomes twice, then let the concentration of B is B′ then
$K_C = [2C]^3 [B']^2 = [C]^3 [B]^2$
or $[B']^2 = \frac{[B]^2}{8}$ or $[B'] = \frac{[B]}{2\sqrt{2}}$
$\therefore$ **(D)**

Problem 8: ***In a reaction at equilibrium 'X' moles of the reactant A decomposes to give 1 mole each of C and D. If the fraction of A decomposed at equilibrium is independent of initial concentration of A then the value 'X' is***

(A) 1 ***(B) 3***
(C) 2 ***(D) 4***

Solution:

$$\begin{array}{llll} xA & \rightleftharpoons & C & + \quad D \\ a & & 0 & \quad 0 \quad \text{at initial} \\ a(1-\alpha) & & \dfrac{a\alpha}{x} & \quad \dfrac{a\alpha}{x} \quad \text{at equilibrium} \end{array}$$

$$K = \frac{[C][D]}{[A]^x} = \frac{\dfrac{\alpha^2 a^2}{x^2 v^2}}{\left[\dfrac{a(1-\alpha)}{V}\right]^x} = \frac{\alpha^2 a^{2-\alpha}}{x^2(1-\alpha)^x V^{2-x}}$$

$\because$ α is independent of A, $\Rightarrow$ 2 – x = 0 or x = 2

$\therefore$ **(C)**

Problem 9: ***If $CuSO_4.5H_2O_{(s)} \rightleftharpoons CuSO_4.3H_2O_{(s)} + 2H_2O_{(l)}$ $K_p = 1.086 \times 10^{-4}$ atm^2 at 25°C. The efflorescent nature of $CuSO_4.5H_2O$ can be noticed when vapour pressure of H_2O in atmosphere is***

(A) $>$7.29 mm ***(B) $<$7.92 mm***
(C) $\geq$7.92 mm ***(D) None***

Solution: An efflorescent salt is one that loses H_2O to atmosphere.

For the reaction

$CuSO_4.5H_2O_{(s)} \rightleftharpoons CuSO_4.3H_2O_{(s)} + 2H_2O_{(l)}$

$K_p = (p'_{H_2O})^2 = 1.086 \times 10^{-4}$

$p'_{H_2O} = 1.042 \times 10^{-2}$ atm = 7.92 mm

$\because$ If p'_{H_2O} at 25°C $<$ 7.92 mm only then, reaction will proceed in forward direction.

$\therefore$ **(B)**

Problem 10: ***In the system, $LaCl_3(s) + H_2O\ (g) + heat \longrightarrow LaClO(s) + 2HCl\ (g)$, equilibrium is established. More water vapour is added to reestablish the equilibrium. The pressure of water vapour is doubled. The factor by which pressure of HCl is changed is***

(A) 2 ***(B)*** $\sqrt{2}$
(C) $\sqrt{3}$ ***(D)*** $\sqrt{5}$

Solution

$LaCl_3(s) + H_2O(g) + heat \rightleftharpoons LaClO(s) + 2HCl(g)$

At eq. P_1 P_2

$$K_P = \frac{P_2^2}{P_1}$$

When $p_{H_2O} = 2P_1$, Let $p_{HCl} = P_2'$

Then $K_P = \frac{P_2'^2}{2P_1}$

$$\frac{P_2'^2}{2P_1} = \frac{P_2^2}{P_1}$$

$$P_2'^2 = 2P_2^2$$

$$P_2' = \sqrt{2}\ P_2$$

therefore pressure of HCl is changed by $\sqrt{\mathbf{2}}$ factor.

$\therefore$ **(B)**

Problem 11: ***The equilibrium constant for the reaction $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is 4×10^{-4} at 200 K. In the presence of a catalyst the equilibrium is attained 10 times faster. Therefore the equilibirum constant in presence of the catalyst at 200 K is***

(A) 4×10^{-3} ***(B) 4×10^{-4}***

(C) 4×10^{-5} ***(D) None***

Solution: Equilibrium constant of a reaction remains unchanged when the same reaction remains unchanged when the same reaction is carried out in presence of a catalyst because the catalyst will catalyse both the forward and backward reactions equally.

$\therefore$**(B)**

Problem 12: ***For the decomposition reaction***

$NH_2COONH_4 (s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

The $K_P = 2.9 \times 10^{-5}$ atm^3. The total pressure of gases at equilibrium when 1 mole of NH_2COONH_4 (s) was taken to start with would be

(A) 0.0194 atm ***(B) 0.0388 atm***

(C) 0.0582 atm ***(D) 0.0766 atm***

Solution: $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

1 2 1

$K_P = 2.9 \times 10^{-5}$ atm^3

If the P is the total pressure at equilibrium

$$K_P = \left(\frac{2p}{3}\right)^2\left(\frac{p}{3}\right) \quad \therefore P^3 = \frac{27 \times 2.9 \times 10^{-5}}{4} = 1.9575$$

$$P = \sqrt[3]{1.9575} = 0.058$$

$\therefore$**(C)**

Problem 13: ***For a reaction $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$, the initial concentration of A and B are equals but the equilibrium concentration of C is twice that of equilibrium concentration of A. Then K is***

(A) 4 (B) 9

(C) 1/4 (D) 1/9

Solution:

	$A_{(g)}$ +	$B_{(g)}$ $\rightleftharpoons$	$C_{(g)}$ +	$D_{(g)}$
Initial conc.	a	a	0	0
Eq. conc.	a - x	a - x	x	x

Given x =2(a-x) or x = $\frac{2a}{3}$

$K_c = \frac{x^2}{(a-x)^2} = \frac{(2a/3)^2}{(a-2a/3)^2} = 4$

$\therefore$ **(A)**

Problem 14: ***The partial pressure of CH_3OH (g), CO(g) and H_2(g) in equilibrium mixture for the reaction CO(g) + $2H_2$(g) $\rightleftharpoons$ CH_3OH(g) are 2.0, 1.0, and 0.1 atm respectively at 427°C. The value of K_p for decomposition of CH_3OH to CO and H_2 is***

(A) 10^2 atm ***(B) 2×10^2 atm^{-1}***
(C) 50 atm^2 ***(D) 5×10^{-3} atm^2***

Solution: $K_p = \frac{P_{CH_3OH}}{P_{H_2}^2 \times P_{CO}} = \frac{2}{1 \times (0.1)^2} = 200$

K_p for reverse reaction will be

$K_P^1 = \frac{1}{K_P} = \frac{1}{200} = 5 \times 10^{-3}$ atm^2

A sample of air consisting of N_2 and O_2 was heated to 2500 K until the equilibrium N_2(g) + O_2(g) $\rightleftharpoons$ 2NO(g) was established with an equilibrium constant $K_c = 2.1 \times 10^{-3}$. At equilibrium, the mole of NO was 1.8. Estimate the initial composition of air in mole fraction of N_2 and O_2.

$\therefore$ **(D)**

Problem 15: ***For reaction A(g) + B(g) $\rightleftharpoons$ AB(g) we start with 2 moles of A and B each. At equilibrium 0.8 moles of AB is formed. Then how much of A changes to AB***

(A) 20% ***(B) 40%***
(C) 60% ***(D) 4%***

Solution:

	A(g)	+	B(g)	$\rightleftharpoons$	AB(g)
Initial moles	2		2		0
At equilibrium	2 - x		2 - x		x = 0.8

$\therefore$ % of A changed to AB = $\frac{0.8}{2} \times 100 = 40\%$

$\therefore$ **(B)**

9. Assignments

9.1 Subjective

LEVEL – I

1. An air sample containing 21:79 of O_2 and N_2 (molar ratio) is heated to 2400°C. If the mole percent of NO at equilibrium is 1.8%. Calculate the K_p of the reaction $N_2 + O_2 \rightleftharpoons 2NO$

2. For the reaction $F_2 \rightleftharpoons 2F$. Calculate the degree of dissociation of fluorine at 4 atm and 1000K, when $K_p = 1.4 \times 10^{-2}$ atm.

3. The equilibrium constant of ester formation of propionic acid with ethyl alcohol is 7.36 at 50°C. Calculate the weight of ethyl propionate in gram existing in an equilibrium mixture when 0.5 mole of propionic acid is heated with 0.5 mole of ethyl alcohol at 50°C.

4. 0.1 mole of ethanol and 0.1 mole of butanoic acid are allowed to react. At equilibrium, the mixture is titrated with 0.85 M NaOH solution and the titre value was 100 ml. Assuming that no ester was hydrolysed by the base, calculate K for the reaction. $C_2H_5OH + C_3H_7COOH \rightleftharpoons C_3H_7COOC_2H_5 + H_2O$

5. To 500 ml of 0.150 M $AgNO_3$ solution we add 500 ml of 1.09 M Fe^{2+} solution and the reaction is allowed to reach equilibrium at 25°C.
$Ag^+ (aq) + Fe^{2+} (aq) \rightleftharpoons Fe^{3+} (aq) + Ag(s)$
For 25 ml. of the solution, 30 ml. of 0.0832 M $KMnO_4$ were required for oxidation under acidic conditions. Calculate K_C for the reaction.

6. When one mole of benzoic acid (C_6H_5COOH) and three moles of ethanol (C_2H_5OH) are mixed and kept at 200°C until equilibrium is reached, it is found that 87% of the acid is consumed by the reaction.
$C_6H_5COOH(l) + C_2H_5OH (l) \rightleftharpoons C_6H_5COOC_2H_5(l) + H_2O(l)$
Find out the percentage of the acid consumed when one mole of the benzoic acid is mixed with four moles of ethanol and treated in the same way.

7. N_2O_4 is 25% dissociated at 37°C and one atmospheric pressure. Calculate (i) K_P and (ii) percent dissociation at 0.1 atmospheric pressure and 37°C.

8. K_p for the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ at 250°C is 0.82. Calculate the degree of dissociation at given temperature under a total pressure of 5 atm. What will be the degree of dissociation if the equilibrium pressure is 10 atm at same temperature.

9. a) Calculate the equilibrium constant for the process
$N_2O_4 \rightleftharpoons 2NO_{2(g)}$
At a temperature 350 K, given that K = 0.14 at 298 K and ΔH = 57.2 kJ

b) The equilibrium constant K_P for the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ is 1.6×10^{-4} atm at 400°C. What will be the equilibrium constant at 500°C if heat of the reaction in this temperature range is –25.14 k cal?

10. a) H_2 and I_2 are mixed at 400°C in a 1.0 L container and when equilibrium established, the following concentrations are present: [HI] = 0.49 M, $[H_2]$ = 0.08 M and $[I_2]$ = 0.06 M. If now an additional 0.3 mol of HI are added, what are the new equilibrium concentrations, when the new equilibrium $H_2 + I_2 \rightleftharpoons 2HI$ is re-established?

b) The equilibrium constant is 0.4 at 300K for the process $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ and it was found that at equilibrium partial pressure of N_2O_4 and NO_2 were 18.9 and 16.5 atm respectively and no. of moles of $N_2O_4 = 7.6 \times 10^{-3}$ mol and $NO_2 = 6.6 \times 10^{-3}$. In what direction would the process move, if any, on adding 1.42×10^{-2} moles of argon

i) If the volume were simultaneously doubled to two litres

ii) If the volume were increased so that total pressure was held constant at 35.4 atm.

iii) If the volume were held constant? (initial amount of N_2O_4 = 1 gm).

LEVEL – II

1. K_C for $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ is 0.04 at 250°C. How many mole of PCl_5 must be added to a 3 litre flask to obtain a Cl_2 concentration of 0.15 M?

2. K_C for $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ is 0.00466 at 298 K. If a one litre container initially contained 0.8 mole of N_2O_4, what would be the concentrations of N_2O_4 and NO_2 at equilibrium? Also calculate equilibrium concentration of N_2O_4 and NO_2 if the volume is halved at the same temperature.

3. A mixture of SO_2, SO_3 and O_2 gas is maintained in a 10 litre flask at a temperature at which K_C = 100 for the reaction
$2SO_2 + O_2 \rightleftharpoons 2SO_3$
a) If the number of moles of SO_2 and SO_3 in the flask are equal, how much O_2 is present?
b) If the number of moles of SO_3 in the flask is twice the number of moles of SO_2. How much O_2 is present?

4. At 800K a reaction mixture contained 0.5 mole of SO_2, 0.12 mole of O_2 and 3 mole of SO_3 at equilibrium. K_C for the equilibrium $2SO_2 + O_2 \rightleftharpoons 2SO_3$ is 833 lit/mol. If the volume of the container is 1 litre. Calculate how much O_2 is to be added at this equilibrium in order to get 5.2 mole of SO_3 at the same temperature.

5. At temperature T, a compound $AB_2(g)$ dissociates according to the reaction, $2AB_2(g) \rightleftharpoons 2AB(g) + B_2(g)$ with degree of dissociation, α, which is small compared to unity. Deduce the expression for α in terms of the equilibrium constant K_P and the total pressure P.

6. Show that K_P for the reaction
$2H_2S(g) \rightleftharpoons 2H_2(g) + S_2(g)$ is given by the expression
$$K_P = \frac{\alpha^3 P}{(2+\alpha)(1-\alpha)^2}$$
Where α is the degree of dissociation and P is the total equilibrium pressure. Calculate K_C of the reaction if α at 298 K and 1 atm pressure is 0.055.

7. The equilibrium constant at 375 K is 2.4 for the process
$SO_2Cl_{2(g)} \rightleftharpoons SO_{2(g)} + Cl_{2(g)}$
Suppose that 6.745g of SO_2Cl_2 is placed in a previously evacuated one litre bulb and the temperature is raised to 375K. What would be the pressure of SO_2Cl_2 if none dissociated? What the partial pressure taking dissociation into account?

8. When 3.06 g of solid NH_4HS is introduced into a two litre evacuated flask at 27°C, 30% of the solid decomposes into gaseous ammonia and hydrogen sulphide. (i) Calculate K_C and K_P for the reaction at 27°C. (ii) What would happen to the equilibrium when more solid NH_4HS is introduced into the flask?

9. a) NO and Br_2 at initial partial pressures of 98.4 and 41.3 torr, respectively, were allowed to react at 300K. At equilibrium the total pressure was 110.5 torr.

Calculate the value of the equilibrium constant and the standard free energy change at 300K for the reaction: $2NO(g) + Br_2(g) \rightleftharpoons 2NOBr(g)$.

b) When 1-pentyne (A) is treated with 4N alcoholic KOH at 175°C, it is converted slowly into an equilibrium mixture of 1.3% 1-pentyne (A), 95.2% 2-pentyne (B) and 3.5% of 1,2-pentadiene (C). The equilibrium was maintained at 175°C. Calculate ΔG^0 for the following equilibria.
B =A ΔG_1^0 = ?, B =C ΔG_2^0 = ?
From the calculated value of ΔG_1^0 and ΔG_2^0 indicate the order of stability of A, B and C. Write a reaction mechanism showing all intermediates leading to A, B and C.

10. a) When PCl_5 is heated it dissociates into PCl_3 and Cl_2. The density of the gas mixture at 200°C and at 250°C is 70.2 and 57.9 respectively. Find the degree of dissociation at 200°C and 250°C.

b) The density of an equilibrium mixture of N_2O_4 and NO_2 at 1 atm. and 348 K is 1.84 g dm^{-3}. Calculate the equilibrium constant of the reaction, $N_2O_4(g) \rightleftharpoons 2NO_2(g)$.

9.2 Objective

SECTION - I

LEVEL – I

1. When KOH is dissolved in water, heat is evolved. If the temperature is raised, the solubility of KOH.
(A) Increases
(B) Decreases
(C) Remains the same
(D) Cannot be predicted

2. For the liquefaction of gas, the favourable conditions are
(A) Low T and high P
(B) Low T and low P
(C) Low T and high P and a catalyst
(D) Low T and catalyst

3. Consider the water gas equilibrium reaction
$C_{(s)} + H_2O_{(g)} \rightleftharpoons CO_{(g)} + H_{2(g)}$
Which of the following statement is true at equilibrium
(A) If the amount of $C_{(s)}$ is increased, less water would be formed
(B) If the amount of $C_{(s)}$ is increased, more CO and H_2 would be formed
(C) If the pressure on the system is increased by halving the volume, more water would be formed.
(D) If the pressure on the system is increased by halving the volume, more CO and H_2 would be formed.

4. A reaction takes place in two steps with equilibrium constants 10^{-2} for slow step and 10^2 for last step. The equilibrium constant of the overall reaction will be
(A) 10^4
(B) 10^{-4}
(C) 1
(D) 10^{-2}

5. For the reaction $PCl_{3(g)} + Cl_{2(g)} \rightleftharpoons PCl_{5(g)}$, the value of K_C at 250°C is 26 mol^{-1}/litre. The value of K_p at this temperature will be
(A) 0.61 atm^{-1}
(B) 0.57 atm^{-1}
(C) 0.85 atm^{-1}
(D) 0.46 atm^{-1}

6. According to Le Chatlier's principle adding heat to a solid and liquid in equilibrium with endothermic nature will cause the
(A) Amount of solid to decrease
(B) Amount of liquid to decrease
(C) Temperature to rise
(D) Temperature to fall

7. An aqueous solution of hydrogen sulphide shows the equilibrium
$H_2S \rightleftharpoons H^+ + HS^-$
If dilute hydrochloric acid is added to an aqueous solution of H_2S, without any change in temperature, then
(A) The equilibrium constant will change
(B) The concentration of HS^- will increase
(C) The concentration of undissociated hydrogen sulphide will decrease
(D) The concentration of HS^- will decrease

8. What would happen to a reversible reaction at equilibrium when temperature is raised, given that its ΔH is positive

(A) More of the products are formed
(B) Less of the products are formed
(C) More of the reactants are formed
(D) It remains in equilibrium

9. Applying the law of mass action to the dissociation of hydrogen Iodide

$2HI(g) \rightleftharpoons H_2 + I_2$

We get the following expression

$$\frac{(a-x)(b-x)}{4x^2} = K$$

Where a is original concentration of H_2, b is original concentration of I_2, x is the number of molecules of H_2 and I_2 reacted with each other. If the pressure is increased in such a reaction then

(A) $K = \frac{(a-x)(b-x)}{4x^2}$
(B) $K > \frac{(a-x)(b-x)}{4x^2}$
(C) $K < \frac{(a-x)(b-x)}{4x^2}$
(D) none of these

10. One mole of ethanol is treated with one mole of ethanoic acid at 25°C. One–fourth of the acid changes into ester at equilibrium. The equilibrium constant for the reaction will be

(A) 1/9
(B) 4/9
(C) 9
(D) 9/4

11. The equilibrium, $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ is attained at 25°C in a closed container and an inert gas He is introduced. Which of the following statements are correct.

(A) concentration of PCl_5, PCl_3 and Cl_2 are changed
(B) more Cl_2 is formed
(C) concentration of PCl_3 is reduced
(D) Nothing happens

12. In which of the following equilibrium, the value of K_P is less than K_C?

(A) $N_2O_4 \rightleftharpoons 2NO_2$
(B) $N_2 + O_2 \rightleftharpoons 2NO$
(C) $N_2 + 3H_2 \rightleftharpoons 2NH_3$
(D) $2SO_2 + O_2 \rightleftharpoons 2SO_3$

13. In the reaction $A_2(g) + 4B_2(g) \rightleftharpoons 2AB_4(g)$, $\Delta H > 0$. The decomposition of AB_4 (g) will be favoured at

(A) low temperature and high pressure
(B) high temperature and low pressure
(C) low temperature and low pressure
(D) high temperature and high pressure

14. For the reaction, $2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$, $K_C = 1.8 \times 10^{-6}$ at 185°C. At 185°C, the value of K_C for the reaction:

$NO(g) + \frac{1}{2}O_2(g) \rightleftharpoons NO_2(g)$ is

(A) 0.9×10^6
(B) 7.5×10^2
(C) 1.95×10^{-3}
(D) 1.95×10^3

15. For the gaseous phase reaction, $2A \rightleftharpoons B + C$, $\Delta H° = –40$ Kcal mol^{-1} which statement is correct for
(A) K is independent of temperature
(B) K increase as temperature decrease
(C) K increase as temperature increases
(D) K varies with addition of A

16. On applying pressure to the equilibrium
Ice $\rightleftharpoons$ water
Which phenomenon will happen
(A) More ice will be formed
(B) More water will be formed
(C) Equilibrium will not be disturbed
(D) Water will evaporate

17. Densities of diamond and graphite are 3.5 and 2.3 grams respectively. Increase of pressure on the equilibrium $C_{diamond} \rightleftharpoons C_{graphite}$
(A) Favours backward reaction
(B) Favours forward reaction
(C) Have no effect
(D) Increase the reaction rate

18. For the reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$ in a vessel after the addition of equal number of mole of N_2 and H_2 equilibrium state is formed which of the following is correct?
(A) $[H_2] = [N_2]$
(B) $[H_2] < [H_2]$
(C) $[H_2] > [N_2]$
(D) $[H_2] > [NH_3]$

19. One mole of $N_2O_{4(g)}$ at 300K is kept in a closed container under one atm. It is heated to 600K when 20% by mass of $N_2O_{4(g)}$ decomposes to $NO_{2(g)}$. The resultant pressure is
(A) 1.2 atm
(B) 2.4 atm
(C) 2.0 atm
(D) 1.0 atm

20. At 30°C, K_p for the dissociation reaction
$SO_{2(g)} \rightleftharpoons SO_{2(g)} + Cl_{2(g)}$
is 2.9×10^{-2} atm. If the total pressure is 1 atm, the degree of dissociation of SO_2Cl_2 is
(A) 87%
(B) 13%
(C) 17%
(D) 29%

21. A vessel at 1000K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted into CO on the addition of graphite. The value of K if the total pressure at equilibrium is 0.8 atm is
(A) 1.8 atm
(B) 3 atm
(C) 0.3 atm
(D) 0.18 atm

22. Vapour density of PCl_5 is 104.16 but when heated at 230°C its vapour density is reduced to 62. The degree of dissociation of PCl_5 at this temperature will be
(A) 6.8%
(B) 68%
(C) 46%
(D) 64%

23. For the reaction $SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$ $K_p = 1.7 \times 10^{12}$ at 20°C and 1 atm pressure. Calculate K_c.
(A) 1.7×10^{12} (B) 0.7×10^{12}
(C) 8.33×10^{12} (D) 1.2×10^{12}

24. For a gaseous equilibrium
$2A(g) \rightleftharpoons 2B(g) + C(g)$, K_p has a value 1.8 at 700°K. What is the value of K_c for the equilibrium $2B(g) + C(g) \rightleftharpoons 2A$ at that temperature
(A) ≈ 0.031 (B) ≈ 32
(C) ≈ 44.4 (D) $\approx 1.3 \times 10^{-3}$

25. For the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ the reaction connecting the degree of dissociation (α) of $N_2O_4(g)$ with its equilibrium constant K_P is
(A) $\alpha = \dfrac{K_P/P}{4+K_P/P}$ (B) $\alpha = \dfrac{K_P}{4+K_P}$
(C) $\alpha = \left[\dfrac{K_P/P}{4+K_P/P}\right]^{1/2}$ (D) $\alpha = \left[\dfrac{K_P}{4+K_P}\right]^{1/2}$

26. In a closed container at 1 atm pressure 2 moles of $SO_2(g)$ and 1 mole of $O_2(g)$ were allowed to react to form $SO_3(g)$ under the influence of acetalyst. Reaction $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ occurred
At equilibrium it was found that 50% of $SO_2(g)$ was converted to $SO_3(g)$. The formal pressure of O_2 (g) at equilibrium will be
(A) 0.66 atm (B) 0.493 atm
(C) 0.33 atm (D) 0.20 atm

27. K_P for a reaction at 25°C is 10 atm. The activation energy for forward and reverse reactions are 12 and 20 kJ / mol respectively. The K_C for the reaction at 40°C will be
(A) 4.33×10^{-1} M (B) 3.33×10^{-2} M
(C) 3.33×10^{-1} M (D) 4.33×10^{-2} M

28. K for the synthesis of HI (g) is 50. The degree of dissociation of HI is
(A) 0.10 (B) 0.14
(C) 0.18 (D) 0.22

29. One mole of $N_2O_4(g)$ at 300 K is kept in a closed container under one atmosphere. It is heated to 600 K when N_2O_4 (g) decomposes to $NO_2(g)$. If the resultant pressure is 2.4 atm, the percentage dissociation by mass of N_2O_4 (g) is
(A) 10% (B) 20%
(C) 30% (D) 40%

30. 40% of a mixture of 0.2 mol of N_2 and 0.6 mol of H_2 react to give NH_3 according to the equation: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ at constant temperature and pressure. Then the ratio of the final volume to the initial volume of gases are
(A) 4:5 (B) 5:4

(C) 7:10 (D) 8:5

LEVEL - II **More than one choice**

1. The dissociation of ammonium carbamate may be represented by the reaction
$NH_4CONH_{2(g)} \rightleftharpoons 2NH_{3(g)} + CO_{2(g)}$
$\Delta H°$ for the forward reaction is negative. The equilibrium will shift from right to left if there is
(A) a decrease in pressure
(B) an increase in temperature
(C) an increase in the concentration of ammonia
(D) an increase in the concentration of carbondioxide

2. The equilibrium of which of the following reactions will not be disturbed by the addition of an inert gas at constant volume
(A) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$
(B) $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$
(C) $CO_{(g)} + 2H_{2(g)} \rightleftharpoons CH_3OH_{(g)}$
(D) $C_{(s)} + H_2O_{(g)} \rightleftharpoons CO_{(g)} + H_{2(g)}$

3. For the chemical reaction 3X (g) + Y(g) $\rightleftharpoons$ X_3Y(g), the amount of X_3Y at equilibrium is affected by
(A) temperature and pressure
(B) temperature only
(C) By adding X(g) and Y(g)
(D) by adding catalyst.

4. For which of the following statements about the reaction quotient, Q, are correct?
(A) the reaction quotient, Q, and the equilibrium constant always have the same numerical value
(B) Q may be >< = K_{eq}
(C) Q (numerical value) varies as reaction proceeds
(d Q = 1 at equilibrium

5. Which of the following statements about the reaction quotient Q, are correct ?
(A) The reaction quotient Q, and the equilibrium constant always have the same numerical value
(B) Q may be > < = K_{eq}.
(C) Q (numerical value) varies as reaction proceeds
(D) Q = 1 at equilibrium

6. For the following equilibrium
$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$
partial pressure of NH_3 will increase:
(A) if NH_3 is added after equilibrium is established
(B) if H_2S is added after equilibrium is established
(C) temperature is increased
(D) volume of the flask is increased

7. Which is/are correct relation(s) for thermodynamic equilibrium constant?
(A) $\Delta G° = -2.303\ RT\ \log K$
(B) $\Delta G = \Delta G° + 2.303\ RT\ \log K$
(C) $E^0 cell = \frac{0.0591}{n} \log K$
(D) $E = E^0 - \frac{0.0591}{n} \log K$

8. In presence of a catalyst, what happens to the chemical equilibrium ?
(A) Energy of activation of the forward and backwards reactions are lowered by same amount

(B) Equilibrium amount is not disturbed
(C) Rates of forward and reverse reaction increase by the same factor
(D) More product is formed.

9. Volume of the flask in which species are transferred is double of the earlier flask. In which of the following cases, equilibrium constant is affected?

(A) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ (B) $N_{2(g)} + O_2 \rightleftharpoons 2NO_{(g)}$

(C) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ (D) $2NO \rightleftharpoons N_{2(g)} + O_{2(g)}$

10. Constant the following equilibrium in a closed container

$N_2O_4\,(g) \rightleftharpoons 2NO_2\,(g)$

At a fixed temperature, the volume of the reaction container is halved. For this change, which of the following statements holds true regarding the equilibrium constant (K_p) and degree of dissociation (α) ?

(A) neither K_p and nor α changes (B) both K_p and α change
(C) α changes (D) K_p does not change

LEVEL - III | Other Engg. Exams

1. AB_2 dissociates as

$AB_2\,(g) \rightleftharpoons AB(g) + B(g)$

When the initial pressure of AB_2 is 600 mm Hg, the total equilibrium pressure is 800 mm Hg. Calculate K for the reaction assuming that the volume of the system remains unchanged.

(A) 50 (B) 100
(C) 166.6 (D) 400.0

2. The state of equilibrium refers to

(A) State of rest (B) Dynamic state
(C) Stationary state (D) State of inertness

3. Consider the two gaseous equilibria involving SO_2 and the corresponding equilibrium constants at 298 K

$SO_2\,(g) + \frac{1}{2}O_2\,(g) \rightleftharpoons SO_3(g)$ . . . . .K_1

$2SO_3\,(g) \rightleftharpoons 2SO_2\,(g) + O_2(g)$ . . . . K_2

The values of the equilibrium constants are related by

(A) $K_2 = K_1$ (B) $K_2 = K_1^2$

(C) $K_2 = \frac{1}{K_1^2}$ (D) $K_2 = \frac{1}{K_1}$

4. The factor which changes equilibrium constant of the reaction

$A_2\,(g) + B_2\,(g) \longrightarrow 2AB\,(g)$ is

(A) Total pressure (B) Amounts of A_2 and B_2
(C) temperature (D) Catalyst

5. The reaction, $SO_2 + Cl_2 \rightleftharpoons SO_2Cl_2$ is exothermic and reversible. A mixture of SO_2 (g), Cl_2(g) and SO_2Cl_2 (*l*) is at equilibrium in a closed container. Now a certain quantity of extra SO_2 is introduced into the container, the volume remaining the same. Which of the following is true ?

(A) The pressure inside the container will not change
(B) The temperature will not change
(C) The temperature will increase
(D) The temperature will decrease.

6. At a certain temperature $2HI \rightleftharpoons H_2 + I_2$, only 50% HI is dissociated at equilibrium. The equilibrium constant is
(A) 0.25 (B) 1.0
(C) 3.0 (D) 0.5

7. If α is the fraction of HI dissociated at equilibrium in the reaction,
$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$, starting with 2 moles of HI, the total number of moles of reactants and products at equilibrium are
(A) $1 + \alpha$ (B) $2 + 2\alpha$
(C) 2 (D) $2 - \alpha$

8. In a reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$, the yield of NH_3 increases on
(A) increasing the temperature
(B) increasing pressure
(C) increasing temperature as well as pressure
(D) decreasing temperature as well as pressure

9. In a reaction $A + 2B \rightleftharpoons 2C$, If 2.0 moles of A, 3.0 moles of B and 2.0 moles of C are placed in a flask of 2L capacity and equilibrium concentration of C is 0.5 mol L^{-1}. The value of equilibrium constant K_c of the reaction is
(A) 0.073 (B) 0.147
(C) 0.05 (D) 0.026

10. In a reaction $A + B \rightleftharpoons C + D$, the initial concentration of A and B were 0.9 mol. dm^{-3} each. At equilibrium the concentration of D was found to be 0.6 mol dm^{-3}. What is the value of equilibrium constant for the reaction ?
(A) 8 (B) 4
(C) 9 (D) 3

11. For the reaction $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$ the partial pressure of CO_2 and CO are 2.0 and 4.0 atm respectively at equilibrium. What is the value of Kp for this reaction ?
(A) 0.5 atm (B) 4.0 atm
(C) 8.0 atm (D) 32.0 atm

12. K_p/K_c for the reaction,
$CO(g) + \frac{1}{2}O_2(g) \rightleftharpoons CO_2(g)$, is
(A) 1 (B) RT
(C) $\frac{1}{\sqrt{RT}}$ (D) $(RT)^{1/2}$

13. The reaction $2SO_2 + 2O_2 \rightleftharpoons 2SO_3$, ΔH = -ve, will be favoured by
(A) high temperature and low pressure
(B) low temperature and high pressure
(C) low temperature and low pressure
(D) high temperature and high pressure

14. A reaction is $A + B \rightleftharpoons C + D$. Initially we start with equal concentrations of A and B. At equilibrium we find that the moles of C is two times of A. What is the equilibrium constant of the reaction ?
(A) 1/4
(B) 1/2
(C) 4
(D) 2

15. 5 moles of SO_2 and 5 moles of O_2 are allowed to react to form SO_3 in a closed vessel. At the equilibrium stage 60% of SO_2 is used up. The total number of moles of SO_2, O_2 and SO_3 in the vessel now is
(A) 3.9
(B) 8.5
(C) 10.5
(D) 10.0

LEVEL - IV

A Matrix-Match Type

1. Match the chemical reaction in equilibrium (in List I) with the pressure dependent of degree of dissociation of the reaction (in List II):

Column - I	**Column - II**
(A) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	(p) $x\ \alpha\sqrt{P}$
(B) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$	(q) $x \propto \sqrt{\frac{1}{P}}$
(C) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$	(r) $x \propto P$
(D) $N_2 + O_2 \rightleftharpoons 2NO$	(s) $x \times P^0$

2.

Column – I	**Column – II**
(A) ΔH = –ve and ΔS = +ve	(p) Reaction feasible only low temperature
(B) ΔH = –ve and ΔS = –ve	(q) Reaction feasible at any temperature
(C) ΔH = +ve and ΔS = –ve	(r) Reaction will not occur at any temperature
(D) ΔH = +ve and ΔS = +ve	(s) Reaction feasible only at high temperature

B Linked Comprehension Type

Comprehension - I

For the reaction $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_2(g)$, K_P(1000 K) = 0.059 and K_P (1100 K) = 0.08. Exactly 10 g of $CaCO_3$ is placed in a 10 litre container at 1000 K. The equilibrium is reached. Assume that air contains 10% by volume of CO_2.

1. The mass of $CaCO_3$ remain unreacted at 1000 K is
(A) 1.485 g
(B) 0.325 g
(C) 9.282 g
(D) 4.56 g

2. The ratio of degree of dissociation of $CaCO_3$ at 1110° K and 1000 K is
(A) 4.256 (B) 1.356
(C) 5.286 (D) 3.456

3. The temperature at which $CaCO_3$ dissociates freely in air is
(A) 415°C (B) 913.2°C
(C) 8.05. 4°C (D) 43°C

4. The value of ΔH is
(A) 6699 cal (B) 2399 cal
(C) 4599 cal (D) 2499 cal

Comprehension - II

Life at high attitudes and hemoglobin production: In the human body, countless chemical equilibria must be maintained to ensure physiological well being. Transport of oxygen by blood depends on the reversible combination of oxygen with haemoglobin. In blood, haemoglobin, oxygen and oxyhaemoglobin are in equilibrium.

$$\text{haemoglobin (Hb)} + O_2 \rightleftharpoons \text{Oxyhaemoglobin } [Hb(O_2)]$$

The equilibrium constant is, $K_c = \dfrac{[Hb(O_2)]}{[Hb][O_2]}$

The pH of blood stream is maintained by a proper balance of H_2CO_3 and $NaHCO_3$ concentrations.

According to Henderson's equation $pH = -\log K_a + \log \dfrac{[salt]}{[acid]}$

An important component of blood is the buffer combination of $H_2PO_4^-$ ion and the HPO_4^- ion. Consider blood with a pH of 7.44. Given $K_{a_1} = 6.9 \times 10^{-3}$, $K_{a_2} = 6.2 \times 10^{-8}$ and $K_{a_3} = 4.8 \times 10^{-13}$.

5. What volume of 5M $NaHCO_3$ solution should be mixed with a 10 mL sample of blood which is 2M in H_2CO_3, in order to maintain a pH of 7.47? K_a for H_2CO_3 in blood is 7.8×10^{-7}.
(A) 78.32 mL (B) 92.08 mL
(C) 68.32 mL (D) None

6. What is the ratio of $\dfrac{[H_2PO_4^-]}{[HPO_4^{2-}]}$?
(A) 0.59 (B) 0.69
(C) 0.79 (D) None

7. What will be the pH, when 25% of the HPO_4^{2-} ions are converted to $H_2PO_4^{2-}$ ion?
(A) 7.16 (B) 8.16
(C) 9.16 (D) None

8. What will be the pH, when 15% of the $H_2PO_4^-$ ions are converted to HPO_4^- ions?
(A) 7.55 (B) 8.55
(C) 9.55 (D) None

Comprehension - III

Let $\Delta G°$ be the difference in free energy of the reaction when all the reactants and products are in the standard state (1 atmospheric pressure and 298K) and K_C and K_P be the thermodynamic equilibrium constant of the reaction. Both are related to each other at temperature T by the following relation:
$\Delta G° = -2.303\ RT \log K_C$ and $\Delta G° = -2.303\ RT \log K_P$ (incase of ideal gas)
This equation represents one of the most important results of thermodynamics and relates to the equilibrium constant of a reaction to a thermodynamic property.
It is sometimes easier to calculate the free energy in a reaction rather than to measure the equilibrium constant.
Standard free energy change can be thermodynamically calculated as
$\Delta G° = \Delta H° - T\Delta S°$
Here $\Delta H°$ = standard enthalpy change
$\Delta S°$ = standard entropy change.

9. Which of the following statement is correct for a reversible process in state of equilibrium
(A) $\Delta G = -2.303\ RT \log K$ (B) $\Delta G = 2.303\ RT \log K$
(C) $\Delta G° = -2.303\ RT \log K$ (D) $\Delta G° = 2.303\ RT \log K$

10. At 490°C, the value of equilibrium constant; K_P is 45.9 the reaction
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
Calculate the value of $\Delta G°$ for the reaction at that temperature
(A) –3.5 kcal (B) 3.5 kcal
(C) 5.79 kcal (D) –5.79 kcal

11. Calculate the equilibrium concentration ratio of C to A if 2.0 mol each of A and B were allowed to come to equilibrium at 300 K
$A + B \rightleftharpoons C + D$ $\Delta G° = 460$ Cal
(A) 1.0 (B) 0.5
(C) 0.8 (D) 0.679

10. Hints for Subjective Problems

LEVEL – I

1. No. of moles of NO: Total no. of moles at equilibrium = 1.8 : 100
2. First calculate K_C.
4. No. of moles of NaOH = No. of moles of acid at equilibrium.
5. $KMnO_4$ reacts with Fe^{+2} at equilibrium.
7. Calculate the partial pressure first.
8. At the same temperature K_P or K_C remains the same.
10. At the same temperature K_P and K_C remains the same.

LEVEL – II

1. No. of moles = molarity $\times$ volume (in litre)
2. $\frac{\text{moles of } O_2}{\text{volume in litre}}$ = concentration of O_2 in molarity
3. On adding O_2 in the equilibrium it will shift towards the forward direction.
4. Here $K_p = p_{CO_2}$ and equilibrium will shift towards the backward directions.
7. Without dissociation the pressure is first because of the no. of moles of SO_2Cl_2.
8. Calculate the no. of moles of NH_4HS first.
10. a) Find $K_{200^\circ C}$ first.

11. Answers

11.1 Subjective

LEVEL – I

1. 2.1×10^{-3}
2. 0.65
3. 36.12
4. 9
5. 3.178
6. 90%
7. i) 0.267 atm; ii) 63.25%
8. 27.5%
9. a) 4.3 b) 1.462×10^{-5} atm
10. a) [HI] = 0.724 M
    b) $[I_2]$ = 0.093

LEVEL – II

1. 2.1 moles
2. moles of O_2 = 0.1, Moles N_2O_4 = 1.557, Moles of NO_2 = 0.086
3. 0.1, 0.4
4. 0.34
5. $\alpha = \left(\frac{2K_P}{P}\right)^{1/3}$
6. 3.71×10^{-6}
7. i) 1.53 atm ii) 0.03 atm

8. 0.19 atm no effect
9. a) 134, 12.2 kJ b) B > C > A
10. a) 0.8 b) 5.3 atm

11.2 Subjective

LEVEL – I

1.	**(B)**	2.	**(A)**
3.	**(C)**	4.	**(C)**
5.	**(A)**	6.	**(A)**
7.	**(D)**	8.	**(A)**
9.	**(A)**	10.	**(A)**
11.	**(D)**	12.	**(C), (D)**
13.	**(C)**	14.	**(B)**
15.	**(C)**	16.	**(B)**
17.	**(A)**	18.	**(B)**
19.	**(B)**	20.	**(C)**
21.	**(A)**	22.	**(B)**
23.	**(C)**	24.	**(C)**
25.	**(C)**	26.	**(D)**
27.	**(C)**	28.	**(D)**
29.	**(B)**	30.	**(A)**

LEVEL – II

1.	**(B), (C), (D)**	2.	**(A), (B), (C), (D)**	3.	**(A), (C)**
4.	**(B), (C)**	5.	**(B), (C)**	6.	**(A), (C), (D)**
7.	**(C), (D)**	8.	**(A), (C)**	9.	**(A), (B), (C)**
10.	**(A), (C)**				

LEVEL - III

1.	**(A)**	2.	**(C)**	3.	**(B)**	4.	**(A)**
5.	**(B)**	6.	**(A)**	7.	**(B, C)**	8.	**(D)**
9.	**(C)**	10.	**(D)**	11.	**(D)**	12	**(B)**
13.	**(B)**	14.	**(B)**	15.	**(B)**		

LEVEL - IV

A	B
Match the following	**Write-up**
1 **(A) – (r)** **(B) – (p)** **(C) – (q)** **(D) – (s)** 2 **(A) – (q)** **(B) – (p)** **(C) – (r)** **(D) – (s)**	1. **(C)** 2. **(B)** 3. **(B)** 4. **(A)** 5. **(A)** 6. **(A)** 7. **(D)** 8. **(A)** 9. **(A)** 10. **(A)** 11. **(D)**